SEMI-ORTHOGONAL SEQUENCES OF $\mathbb{P}$-LIKE OBJECTS, GLUED T-STRUCTURES, AND STRATIFIED ALGEBRAS

Abstract. Bodzenta and Bondal characterised highest weight categories as hearts of t-structures glued along full exceptional sequences, under a vanishing condition on negative extensions within the sequence and within its left dual. We investigate whether exceptional objects can be replaced by $\mathbb{P}^n[k]$-like objects, i.e. objects whose graded endomorphism algebra is $\Bbbk[t]/(t^{n+1})$ with $\deg t = k$, with the aim of producing stratified algebras.

We prove a gluing theorem for admissible filtrations of triangulated categories whose strata carry bounded t-structures with local hearts. Under a vanishing condition on degree-two morphisms in each stratum, the glued heart is the module category of a standardly stratified algebra. Its standard and proper costandard modules are the extensions by zero of the stratum projectives and of the stratum simples, respectively. The ambient category is the bounded derived category of the heart precisely when all positive-degree morphisms from stratum projectives to stratum simples vanish.

Applied to semi-orthogonal sequences of $\mathbb{P}^n[k]$-like objects, this shows that the outcome depends on k. For $k \geq 3$ the glued heart is highest weight. For $k = 1$ and $n \geq 3$ it is standardly stratified, and the endomorphism algebras of all non-exceptional standard objects are the dual numbers. For $k = n = 1$ the glued heart is not a module category. In all cases with $n \geq 1$ the ambient category is not the derived category of the heart.

Conversely, the standard modules of a standardly stratified algebra form a semi-orthogonal sequence of objects without self-extensions in non-zero degrees and with local endomorphism algebras; these objects are $\mathbb{P}^n[0]$-like when their endomorphism algebras are truncated polynomial algebras. Their Hom-dual objects, the proper costandard modules, have infinite projective dimension unless the algebra is quasi-hereditary.

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1. Introduction

1.1. **Background.** Highest weight categories, introduced by Cline, Parshall and Scott [CPS88], are precisely the module categories of quasi-hereditary algebras. Their bounded derived categories carry two distinguished full exceptional sequences: the standard objects $(\Delta(\lambda_1), \dots, \Delta(\lambda_n))$, and the costandard objects, which form the left dual sequence $(\nabla(\lambda_n), \dots, \nabla(\lambda_1))$.

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Bodzenta and Bondal [BB26] proved that this observation characterises highest weight categories, in two ways:

- The heart of the t-structure glued along a full exceptional sequence is highest weight, with the given sequence and its left dual as standard and costandard objects, if and only if neither sequence has negative extensions (Theorem 2.20).
- Highest weight structures on an abelian category $\mathcal{A}$ correspond bijectively to pairs of dual full exceptional sequences in $\mathcal{D}^b(\mathcal{A})$ consisting of objects of $\mathcal{A}$ (Theorem 2.22).

Intuition. A full exceptional sequence is the same as an admissible filtration of $\mathcal{D}$ whose subquotients (the *strata*) are all equivalent to $\mathcal{D}^b(\Bbbk)$. Each stratum has, up to shift, a unique bounded t-structure. Its heart is $\operatorname{mod}\Bbbk$, whose unique simple object $\Bbbk$ is at the same time projective. The standard object E_s is the extension by zero $j_{s!}(\Bbbk)$ of this object, and the costandard object F_s is its extension $j_{s*}(\Bbbk)$. Highest weight categories are exactly what one obtains by gluing such trivial strata, provided the objects E_s, F_s land in the glued heart.

1.2. **The question.** *Standardly stratified algebras* and *properly stratified algebras* [CPS96, Dla96, AHLU00] generalise quasi-hereditary algebras. The endomorphism algebras $B_i = \operatorname{End}(\Delta(i))$ of the standard modules are allowed to be arbitrary local algebras instead of $\Bbbk$.

On the triangulated side, exceptional objects are the case $n = 0$ of $\mathbb{P}^n[k]$*-like objects* [HT06, Kru18, HK19]: objects P with $\operatorname{End}^\bullet(P) \cong \Bbbk[t]/(t^{n+1})$ and $\deg t = k$. Bonfert [Bonf26] has classified the t-structures and weight structures on the triangulated category $\langle P\rangle$ generated by such an object. For $n > 0$ there is, up to shift, a unique non-trivial t-structure if $k > 0$, and a unique non-trivial weight structure if $k \leq 0$ (Theorem 3.6). This is exactly the local input that a gluing construction requires. We address the following question.

Question 1.1. *Is there a generalisation of (dual pairs of) exceptional sequences in which exceptional objects are replaced by $\mathbb{P}^n[k]$-like objects, and which recovers stratified algebras in the same way as dual pairs of exceptional sequences recover quasi-hereditary algebras?*

1.3. **Results.** The basic observation is that the glued heart sees each stratum only through the heart of that stratum, together with two distinguished objects of it: its simple object and the projective cover of that simple object. The self-extensions of a $\mathbb{P}^n[k]$-like object therefore matter only through the following two pieces of data:

(i) the heart $\mathcal{H}_P$ of the natural t-structure on $\langle P\rangle$, which is computed in Proposition 3.14;
(ii) the morphisms of degree 1 and 2 between the projective and simple objects of $\mathcal{H}_P$.

Table 1 summarises the outcome.

Theorem A (Theorem 4.8)**.** *Let $\mathcal{D}$ be a Hom-finite triangulated category with an admissible filtration $0 = \mathcal{D}_0 \subset \mathcal{D}_1 \subset \cdots \subset \mathcal{D}_m = \mathcal{D}$. Suppose that each stratum $\mathcal{T}_s = \mathcal{D}_s/\mathcal{D}_{s-1}$ carries a bounded t-structure whose heart is a finite length category with a unique simple object S_s, and that S_s has a projective cover Π_s. Put $\Delta_s = j_{s!}\Pi_s$ and $\overline{\nabla}_s = j_{s*}S_s$, and let $\mathcal{A}$ be the heart of the glued t-structure. Assume that $\Delta_s, \overline{\nabla}_s \in \mathcal{A}$ for all s; this is equivalent to the vanishing of negative extensions among the Δ_s and among the $\overline{\nabla}_s$. Assume moreover that $\operatorname{Hom}_{\mathcal{T}_s}(\Pi_s, S_s[2]) = 0$ for all s. Then:*

(1) *$\mathcal{A} \simeq \operatorname{mod}\Lambda$ for a standardly stratified algebra Λ, with standard modules Δ_s, proper costandard modules $\overline{\nabla}_s$, and $\operatorname{End}(\Delta_s) \cong \operatorname{End}(\Pi_s)$.*
(2) *If $\mathcal{D}$ is algebraic, the realisation functor $\mathcal{D}^b(\mathcal{A}) \to \mathcal{D}$ is an equivalence if and only if $\operatorname{Hom}_{\mathcal{T}_s}(\Pi_s, S_s[l]) = 0$ for all s and all $l \geq 1$.*

For $\mathcal{T}_s = \mathcal{D}^b(\Bbbk)$ this recovers the Bodzenta–Bondal criterion (Theorem 2.20).

stratum	heart of the stratum	glued heart $\mathcal{A}$	$\mathcal{D}^b(\mathcal{A}) \simeq \mathcal{D}$?
$n = 0$ (exceptional)	$\operatorname{mod}\Bbbk$	highest weight [BB26]	yes
$\mathbb{P}^n[k]$, $k \geq 3$	$\operatorname{mod}\Bbbk$	highest weight (Thm. B)	no
$\mathbb{P}^n[k]$, $k = 2$	$\operatorname{mod}\Bbbk$	open (Question 5.7)	no*
$\mathbb{P}^n[1]$, $n \geq 3$	$\operatorname{mod}\Bbbk[x]/(x^2)$	std. stratified (Thm. B)	no
$\mathbb{P}^2[1]$	$\operatorname{mod}\Bbbk[x]/(x^2)$	open (Question 5.7)	no*
$\mathbb{P}^1[1]$	nilpotent $\Bbbk[x]$-modules	not Deligne finite (Thm. B)	—
$\mathbb{P}^n[0]$	no t-structure [Bonf26]	— (but see Thm. C)	—
$\mathcal{D}^b(\operatorname{mod} B)$, B local	$\operatorname{mod} B$	std. stratified (Thm. A)	yes

TABLE 1. Strata of a semi-orthogonal filtration and the resulting glued heart. In the rows with $\mathbb{P}^n[k]$ we assume $n \geq 1$. "Std. stratified" means standardly stratified; "no t-structure" means no non-trivial t-structure; the asterisk means "no, whenever $\mathcal{A}$ is a module category of the expected type".

Theorem B (Theorem 5.4). *Let $(P_1, \dots, P_m)$ be a full admissible semi-orthogonal sequence of objects such that each P_s is $\mathbb{P}^{n_s}[k_s]$-like with $k_s \geq 1$. Glue the t-structures whose hearts are $\operatorname{Filt}(P_s)$; by [Bonf26], every non-trivial t-structure on $\langle P_s \rangle$ is a shift of this one. Assume that $E_s := P_s$ and $F_s := j_{s*}P_s$ lie in the glued heart $\mathcal{A}$; equivalently, neither sequence has negative extensions.*

1. *If for every s either $n_s = 0$ or $k_s \geq 3$, then $\mathcal{A}$ is a highest weight category with standard objects E_s and costandard objects F_s.*
2. *If for every s either $n_s = 0$, or $k_s \geq 3$, or ($k_s = 1$ and $n_s \geq 3$), then $\mathcal{A}$ is the module category of a standardly stratified algebra. Its standard objects have endomorphism algebra $\Bbbk[x]/(x^2)$ for $k_s = 1$, and its proper standard and proper costandard objects are E_s and F_s.*
3. *If $k_s = n_s = 1$ for some s, then the simple object L_s of $\mathcal{A}$ has no projective cover.*

In (1) and (2), $\mathcal{D}$ is equivalent to $\mathcal{D}^b(\mathcal{A})$ via realisation if and only if all $n_s = 0$.

Theorem C (Theorem 6.3). *Let Λ be a standardly stratified algebra with $B_i = \operatorname{End}(\Delta(i))$.*

1. *The standard modules form a full semi-orthogonal sequence in $K^b(\operatorname{proj}\Lambda)$ of objects without self-extensions in non-zero degrees and with local endomorphism algebras B_i. In particular $\Delta(i)$ is $\mathbb{P}^n[0]$-like if $B_i \cong \Bbbk[t]/(t^{n+1})$.*
2. *$\operatorname{Hom}^\bullet(\Delta(i), \overline{\nabla}(i)) \cong \Bbbk$ and $\operatorname{Hom}^\bullet(\Delta(j), \overline{\nabla}(i)) = 0$ for $j > i$.*
3. *If $B_i \neq \Bbbk$, then $\overline{\nabla}(i) \notin K^b(\operatorname{proj}\Lambda)$.*
4. *In particular, $\operatorname{gl.dim}\Lambda < \infty$ if and only if Λ is quasi-hereditary.*

Answer to Question 1.1. Taken together, the results give the following answer.

- To recover genuinely stratified (non-quasi-hereditary) algebras together with their derived categories, the right generalisation of an exceptional object is a *local-exceptional* object: one with no self-extensions in non-zero degrees and a local endomorphism algebra. $\mathbb{P}^n[0]$-like objects are an example.
- Such an object must be paired not with a mutation-dual object, but with the simple object of its stratum. That simple object lies outside the triangulated category generated by the sequence (Theorem C), so the ambient category has to be of "$\mathcal{D}^b$-type" rather than "perfect type"; Theorem A is formulated accordingly.

- For $\mathbb{P}^n[k]$-like objects with $k \geq 1$ and n finite, the stratum heart is semisimple unless $k = 1$. Gluing therefore yields highest weight categories for $k \geq 3$, with an ambient category that is not their derived category.
- The case $k = 1$, $n \geq 3$ does produce stratified hearts, but only with dual-number endomorphism algebras, and again the ambient category is not the derived category of the heart.
- The geometrically most important case $k = 2$, which contains the Huybrechts–Thomas $\mathbb{P}^n$-objects, is not decided by our argument.

Sources and status of the statements. Every statement attributed to the literature is quoted with its statement, in our notation; we have tried to state the cited results exactly as they are used. All statements without a citation are proved here, except those labelled *Question*. From Bonfert's preprint [Bonf26] we use the formality and classification results quoted in Section 3. Our computation of the hearts of the categories generated by $\mathbb{P}^n[k]$-like objects (Proposition 3.14) overlaps with [Bonf26, Lemmas 3.11 and 4.15]; we include proofs in the form needed for the gluing arguments.

Organisation. Section 2 collects definitions and the results we use: t-structures, gluing, simple-minded collections, exceptional sequences, the Bodzenta–Bondal theorems, and stratified algebras. Section 3 defines $\mathbb{P}^n[k]$-like objects and computes the hearts of the categories they generate. Section 4 proves Theorem A. Section 5 treats sequences of $\mathbb{P}^n[k]$-like objects (Theorem B). Section 6 relates standardly stratified algebras to local-exceptional sequences (Theorem C) and discusses a converse. Section 7 lists open questions.

2. Preliminaries

2.1. **Conventions.** Throughout, $\Bbbk$ is an algebraically closed field. Triangulated categories are $\Bbbk$-linear, *Hom-finite* (all $\mathrm{Hom}(X,Y)$ are finite-dimensional) and idempotent complete; in particular they are Krull–Schmidt. We use the following notation.

- For objects X, Y we write $\mathrm{Hom}^l(X,Y) := \mathrm{Hom}(X, Y[l])$ and $\mathrm{Hom}^\bullet(X,Y) := \bigoplus_{l\in\mathbb{Z}} \mathrm{Hom}^l(X,Y)$. Thus $\mathrm{End}^\bullet(X)$ is a graded algebra under composition.
- For a class $\mathcal{S}$ of objects, $\langle \mathcal{S} \rangle$ is the smallest thick subcategory (triangulated and closed under direct summands) containing $\mathcal{S}$.
- $\mathrm{Filt}(\mathcal{S})$ is the smallest full subcategory containing 0 and $\mathcal{S}$ and closed under extensions: if $X \to Y \to Z \to X[1]$ is a triangle with $X, Z \in \mathrm{Filt}(\mathcal{S})$, then $Y \in \mathrm{Filt}(\mathcal{S})$.
- We put $\mathcal{S}^\perp = \{Y : \mathrm{Hom}^\bullet(S,Y) = 0 \ \forall S \in \mathcal{S}\}$, and define ${}^\perp\mathcal{S}$ dually.

A triangulated category is *algebraic* if it is equivalent to the homotopy category H^0 of a pre-triangulated dg category (equivalently, to the stable category of a Frobenius exact category).

Modules are finite-dimensional right modules, and $\mathrm{mod}\,\Lambda$ denotes their category; $K^b(\mathrm{proj}\,\Lambda) \subset \mathcal{D}^b(\mathrm{mod}\,\Lambda)$ is the subcategory of perfect complexes. In an abelian length category, $[X : L]$ denotes the composition multiplicity of a simple object L in X. For a set Θ of objects, $\mathcal{F}(\Theta)$ is the full subcategory of objects admitting a finite filtration with subquotients isomorphic to objects in Θ.

2.2. **t-structures and hearts.**

Definition 2.1 ([BBD82, Déf. 1.3.1])**.** A *t-structure* on $\mathcal{D}$ is a pair $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ of strictly full subcategories such that, with $\mathcal{D}^{\leq n} = \mathcal{D}^{\leq 0}[-n]$ and $\mathcal{D}^{\geq n} = \mathcal{D}^{\geq 0}[-n]$:

(1) $\mathcal{D}^{\leq 0}[1] \subset \mathcal{D}^{\leq 0}$ and $\mathcal{D}^{\geq 0}[-1] \subset \mathcal{D}^{\geq 0}$;
(2) $\mathrm{Hom}(X,Y) = 0$ for $X \in \mathcal{D}^{\leq 0}$ and $Y \in \mathcal{D}^{\geq 1}$;

(3) every X fits into a triangle $X' \to X \to X'' \to X'[1]$ with $X' \in \mathcal{D}^{\leq 0}$ and $X'' \in \mathcal{D}^{\geq 1}$.

The *heart* $\mathcal{A} = \mathcal{D}^{\leq 0} \cap \mathcal{D}^{\geq 0}$ is an abelian category, with cohomology functors $H^l \colon \mathcal{D} \to \mathcal{A}$. The t-structure is *bounded* if every object lies in $\mathcal{D}^{\leq a} \cap \mathcal{D}^{\geq b}$ for some a, b. It has *length heart* if $\mathcal{A}$ is a finite length category with finitely many isomorphism classes of simple objects.

Short exact sequences in $\mathcal{A}$ are exactly the triangles in $\mathcal{D}$ all of whose vertices lie in $\mathcal{A}$. Consequently $\mathcal{A}$ is closed under extensions in $\mathcal{D}$.

Proposition 2.2 ([BBD82, Rem. 3.1.17]). *Let $\mathcal{A}$ be the heart of a t-structure on $\mathcal{D}$, and let $X, Y \in \mathcal{A}$. The natural maps $\operatorname{Ext}^l_{\mathcal{A}}(X, Y) \to \operatorname{Hom}_{\mathcal{D}}(X, Y[l])$, which send a Yoneda extension to the composite of the corresponding connecting morphisms, are bijective for $l \leq 1$ and injective for $l = 2$. They are compatible with Yoneda product and composition.*

Theorem 2.3 (Realisation; [BBD82, §3.1], [Bei87], [Let24]). *Let $\mathcal{D}$ be algebraic (more generally, the base of a filtered triangulated category), and let $\mathcal{A}$ be the heart of a bounded t-structure on $\mathcal{D}$.*

(1) *There is an exact functor* real$\colon \mathcal{D}^b(\mathcal{A}) \to \mathcal{D}$ *which restricts to the inclusion on $\mathcal{A}$ and induces the maps of Proposition 2.2 on extension groups.*
(2) real *is fully faithful if and only if the following* effaceability *condition holds: for all $X, Y \in \mathcal{A}$, all $n \geq 2$ and all $f \in \operatorname{Hom}_{\mathcal{D}}(X, Y[n])$, there is an epimorphism $X' \twoheadrightarrow X$ in $\mathcal{A}$ such that the composite $X' \to X \xrightarrow{f} Y[n]$ vanishes.*
(3) *In this case* real *is an equivalence, since $\mathcal{D}$ is generated by $\mathcal{A}$ under extensions and shifts.*

Remark 2.4. In [BBD82, Prop. 3.1.16] the effaceability condition is formulated with monomorphisms $Y \hookrightarrow Y'$. The version with epimorphisms follows by applying the statement to $\mathcal{D}^{\mathrm{op}}$. The existence of real for algebraic triangulated categories is established, for instance, in [Let24].

2.3. Simple-minded collections, silting objects and co-t-structures.

Definition 2.5. A finite set $\{S_1, \dots, S_r\}$ of objects of $\mathcal{D}$ is a *simple-minded collection* if

(SM1) $\operatorname{Hom}(S_i, S_j[l]) = 0$ for all i, j and all $l < 0$;
(SM2) $\operatorname{Hom}(S_i, S_j) = \delta_{ij}\Bbbk$;
(SM3) $\langle S_1, \dots, S_r \rangle = \mathcal{D}$.

An object T is *silting* if $\operatorname{Hom}(T, T[l]) = 0$ for all $l > 0$ and $\langle T \rangle = \mathcal{D}$. It is *tilting* if moreover $\operatorname{Hom}(T, T[l]) = 0$ for all $l < 0$.

Intuition. Simple-minded collections behave like the simple modules of a finite-dimensional algebra, and silting objects behave like its projective modules. The former determine t-structures, the latter co-t-structures (weight structures). For $\mathcal{D}^b(\operatorname{mod}\Lambda)$ the two notions correspond to each other under the bijections of Koenig and Yang (Theorem 2.9). This "simple versus projective" dichotomy is the mechanism behind Theorem 3.4 and behind all results below.

Theorem 2.6 ([AN09]; see also [KY14, §5]). *Let $\{S_1, \dots, S_r\}$ be a simple-minded collection in a Hom-finite Krull–Schmidt triangulated category $\mathcal{D}$. Put*

$$\mathcal{D}^{\leq 0} := \operatorname{Filt}\{S_i[l] : 1 \leq i \leq r,\ l \geq 0\}, \qquad \mathcal{D}^{\geq 0} := \operatorname{Filt}\{S_i[l] : 1 \leq i \leq r,\ l \leq 0\}.$$

Then $(\mathcal{D}^{\leq 0}, \mathcal{D}^{\geq 0})$ is a bounded t-structure on $\mathcal{D}$. Its heart is $\operatorname{Filt}\{S_1, \dots, S_r\}$, a length category whose simple objects are exactly $S_1, \dots, S_r$.

Definition 2.7 ([Bon10, Pau08]). A *co-t-structure* (or *weight structure*) on $\mathcal{D}$ is a pair $(\mathcal{U}, \mathcal{V})$ of full subcategories with the following properties: both are closed under direct summands;

$\mathcal{U}[-1] \subset \mathcal{U}$ and $\mathcal{V}[1] \subset \mathcal{V}$; $\mathrm{Hom}(\mathcal{U}, \mathcal{V}) = 0$; and every object X fits into a triangle $U \to X \to V \to U[1]$ with $U \in \mathcal{U}$ and $V \in \mathcal{V}$. Its *co-heart* is $\mathcal{U} \cap \mathcal{V}[-1]$. It is *bounded* if $\bigcup_l \mathcal{U}[l] = \mathcal{D} = \bigcup_l \mathcal{V}[l]$.

Theorem 2.8 ([Bon10, AI12])**.** *Let T be a silting object of $\mathcal{D}$. Let $\mathcal{U}$ (resp. $\mathcal{V}$) be the smallest full subcategory closed under extensions and direct summands containing all $T[l]$ with $l \leq 0$ (resp. $l \geq 1$). Then $(\mathcal{U}, \mathcal{V})$ is a bounded co-t-structure with co-heart* $\operatorname{add} T$.

Theorem 2.9 (Koenig–Yang [KY14, Thm. 6.1])**.** *For a finite-dimensional algebra Λ there are bijections between:*

1. *equivalence classes of silting objects in $K^b(\operatorname{proj}\Lambda)$;*
2. *equivalence classes of simple-minded collections in $\mathcal{D}^b(\operatorname{mod}\Lambda)$;*
3. *bounded t-structures on $\mathcal{D}^b(\operatorname{mod}\Lambda)$ with length heart;*
4. *bounded co-t-structures on $K^b(\operatorname{proj}\Lambda)$.*

Theorem 2.10 (Keller [Kel94])**.** *Let $\mathcal{D}$ be an algebraic idempotent complete triangulated category, and let $T \in \mathcal{D}$ be a tilting object. Then there is a triangle equivalence $\mathcal{D} \simeq K^b(\operatorname{proj}\operatorname{End}(T))$ sending T to the free module of rank one.*

2.4. Recollements and gluing.

Definition 2.11. A full triangulated subcategory $\mathcal{N} \subset \mathcal{M}$ is *admissible* if the inclusion $i_*\colon \mathcal{N} \to \mathcal{M}$ has a left adjoint i^* and a right adjoint $i^!$. Put $\mathcal{Q} = \mathcal{M}/\mathcal{N}$ (Verdier quotient), and let $j^*\colon \mathcal{M} \to \mathcal{Q}$ be the quotient functor. It has a left adjoint $j_!$ and a right adjoint j_*, both fully faithful, which identify $\mathcal{Q}$ with ${}^\perp\mathcal{N}$ and with $\mathcal{N}^\perp$ respectively. We obtain a *recollement*

$$\mathcal{N} \underset{i^!}{\overset{i^*}{\underset{\longleftarrow}{\overset{\longleftarrow}{\longrightarrow}}}} \mathcal{M} \underset{j_*}{\overset{j_!}{\underset{\longleftarrow}{\overset{\longleftarrow}{\longrightarrow}}}} \mathcal{Q}$$

with $i^*j_! = 0$, $i^!j_* = 0$, $j^*i_* = 0$, $j^*j_! \cong \mathrm{id} \cong j^*j_*$, and functorial triangles

$$j_!j^*X \to X \to i_*i^*X \to, \qquad i_*i^!X \to X \to j_*j^*X \to .$$

Applying $i^!$ to the first triangle for $X = j_*Y$ gives $i^!j_!Y \cong i^*j_*Y[-1]$.

Theorem 2.12 (Gluing [BBD82, Thm. 1.4.10 and §1.4])**.** *In a recollement as above, let $\mathcal{N}$ and $\mathcal{Q}$ carry t-structures. Then*

$$\mathcal{M}^{\leq 0} = \{X : j^*X \in \mathcal{Q}^{\leq 0},\ i^*X \in \mathcal{N}^{\leq 0}\},$$
$$\mathcal{M}^{\geq 0} = \{X : j^*X \in \mathcal{Q}^{\geq 0},\ i^!X \in \mathcal{N}^{\geq 0}\}$$

is a t-structure on $\mathcal{M}$, the glued t-structure*. With respect to it, i_* and j^* are t-exact, $j_!$ and i^* are right t-exact, and j_* and $i^!$ are left t-exact. If the given t-structures are bounded, so is the glued one.*

Let $\mathcal{A}_\mathcal{N}$, $\mathcal{A}$ and $\mathcal{A}_\mathcal{Q}$ be the three hearts. Write ${}^pH^0$ for the cohomology functors of the glued t-structure, and set ${}^pj_! := {}^pH^0 \circ j_!$ and ${}^pj_* := {}^pH^0 \circ j_*$ on $\mathcal{A}_\mathcal{Q}$. For $Y \in \mathcal{A}_\mathcal{Q}$ define the *intermediate extension*

$$j_{!*}Y := \operatorname{Im}\left({}^pj_!Y \to {}^pj_*Y\right).$$

Proposition 2.13 ([BBD82, §1.4])**.** *In the situation of Theorem 2.12:*

1. *i_* identifies $\mathcal{A}_\mathcal{N}$ with the Serre subcategory $\{X \in \mathcal{A} : j^*X = 0\}$ of $\mathcal{A}$; this subcategory is closed under subobjects, quotients and extensions.*

(2) *For $Y \in \mathcal{A}_\mathcal{Q}$ we have $j^* {}^p j_! Y \cong Y \cong j^* {}^p j_* Y \cong j^* j_{!*} Y$. The object ${}^p j_! Y$ has no non-zero quotient in $i_* \mathcal{A}_\mathcal{N}$, and ${}^p j_* Y$ has no non-zero subobject in $i_* \mathcal{A}_\mathcal{N}$. Hence $j_{!*} Y$ has neither. The kernel of ${}^p j_! Y \twoheadrightarrow j_{!*} Y$ and the cokernel of $j_{!*} Y \hookrightarrow {}^p j_* Y$ lie in $i_* \mathcal{A}_\mathcal{N}$.*

(3) *The simple objects of $\mathcal{A}$ are the $i_* S$ with S simple in $\mathcal{A}_\mathcal{N}$, and the $j_{!*} S$ with S simple in $\mathcal{A}_\mathcal{Q}$.*

(4) *If $\mathcal{A}_\mathcal{N}$ and $\mathcal{A}_\mathcal{Q}$ are length categories, so is $\mathcal{A}$.*

Comments on the proof. (1) and (3) are proved in [BBD82, §1.4] (see in particular 1.4.22–1.4.26 there, where $j_{!*}$ is introduced and the simple objects of $\mathcal{A}$ are determined).

For (2): j^* is t-exact, so $j^* {}^p j_! Y = H^0(j^* j_! Y) = Y$, and similarly for j_*. Moreover $\mathrm{Hom}_\mathcal{A}({}^p j_! Y, i_* Z) = \mathrm{Hom}(j_! Y, i_* Z) = \mathrm{Hom}(Y, j^* i_* Z) = 0$, and dually for ${}^p j_* Y$. The map ${}^p j_! Y \to {}^p j_* Y$ becomes an isomorphism after applying the exact functor $j^*|_\mathcal{A}$, so its kernel and cokernel lie in $i_* \mathcal{A}_\mathcal{N}$ by (1).

For (4): every $X \in \mathcal{A}$ sits in an exact sequence

$$0 \to i_* H^{-1}(i^* X) \to {}^p j_! j^* X \to X \to i_* H^0(i^* X) \to 0,$$

obtained from the long exact cohomology sequence of $j_! j^* X \to X \to i_* i^* X$. The functor ${}^p j_!$ is right exact. By (2), ${}^p j_!$ of a simple object has finite length, hence so does ${}^p j_!$ of any object of finite length. Thus X has finite length. □

Definition 2.14. An *admissible filtration* of $\mathcal{D}$ is a chain $0 = \mathcal{D}_0 \subset \mathcal{D}_1 \subset \cdots \subset \mathcal{D}_m = \mathcal{D}$ of thick subcategories such that $\mathcal{D}_{s-1}$ is admissible in $\mathcal{D}_s$ for $1 \le s \le m$.

- Its *strata* are $\mathcal{T}_s := \mathcal{D}_s / \mathcal{D}_{s-1}$. We write i_s, j_s for the functors of the recollement $(\mathcal{D}_{s-1}, \mathcal{D}_s, \mathcal{T}_s)$.
- By abuse of notation we regard $j_{s!}$ and j_{s*} as functors $\mathcal{T}_s \to \mathcal{D}$, by composing with the inclusion $\mathcal{D}_s \subset \mathcal{D}$.
- Given t-structures on all $\mathcal{T}_s$, the *glued t-structure* on $\mathcal{D}$ is defined inductively: the glued t-structure on $\mathcal{D}_s$ is obtained from that on $\mathcal{D}_{s-1}$ and the one on $\mathcal{T}_s$ by Theorem 2.12. (For $s = 1$ we have $\mathcal{D}_1 = \mathcal{T}_1$.)

By Theorem 2.12, the inclusions $\mathcal{D}_s \subset \mathcal{D}$ are t-exact. Hence $\mathcal{A}_s := \mathcal{A} \cap \mathcal{D}_s$ is the heart of $\mathcal{D}_s$, and it is a Serre subcategory of $\mathcal{A}$.

2.5. **Exceptional sequences.**

Definition 2.15 ([Bon89]; conventions of [BB26])**.** An object E is *exceptional* if $\mathrm{End}^\bullet(E) = \Bbbk$. A sequence $(E_1, \dots, E_n)$ of exceptional objects is an *exceptional sequence* if $\mathrm{Hom}^\bullet(E_j, E_i) = 0$ for $j > i$. It is *full* if $\langle E_1, \dots, E_n \rangle = \mathcal{D}$.

For an exceptional pair (E, F), the left mutation $L_E F$ is defined by the triangle $\mathrm{Hom}^\bullet(E, F) \otimes E \to F \to L_E F$. The *left dual* of $(E_1, \dots, E_n)$ is $(F_n, \dots, F_1)$, where $F_1 = E_1$ and $F_i = L_{E_1} \cdots L_{E_{i-1}} E_i$.

A full exceptional sequence gives the admissible filtration $\mathcal{D}_s = \langle E_1, \dots, E_s \rangle$, with all strata equivalent to $\mathcal{D}^b(\Bbbk)$. In the notation of Definition 2.14 one has $E_s = j_{s!}\Bbbk$ and $F_s = j_{s*}\Bbbk$ [BB26, §2.4]. If the sequence is full, the left dual is characterised by the *Hom-duality*

$$\mathrm{Hom}^\bullet(E_i, F_j) \cong \delta_{ij} \Bbbk \quad \text{(concentrated in degree 0).} \tag{1}$$

The *t-structure glued along* $(E_1, \dots, E_n)$ is the one obtained by gluing the standard t-structures on $\mathcal{D}^b(\Bbbk)$.

Theorem 2.16 ([BB26, Lemma 2.4])**.** *Let $(E_1, \dots, E_n)$ be a full exceptional sequence with left dual $(F_n, \dots, F_1)$. The t-structure glued along it is*

$$\mathcal{D}^{\le 0} = \{X : \mathrm{Hom}(X, F_s[l]) = 0 \ \forall s, \ \forall l < 0\}, \qquad \mathcal{D}^{\ge 0} = \{X : \mathrm{Hom}(E_s, X[l]) = 0 \ \forall s, \ \forall l < 0\}.$$

2.6. **Highest weight categories and the Bodzenta–Bondal theorems.**

Definition 2.17 ([CPS88]; cf. [BB26, §2.1]). A $\Bbbk$-linear abelian category is *Deligne finite* if it is a Hom-finite length category with finitely many simple objects and a projective generator; equivalently, it is equivalent to $\operatorname{mod}\Lambda$ for a finite-dimensional algebra Λ.

Let $\mathcal{A}$ be Deligne finite, with simple objects $L(1),\dots,L(m)$ indexed by the totally ordered set $\{1<\cdots<m\}$, projective covers $P(i)$ and injective hulls $I(i)$. Then $\mathcal{A}$ is *highest weight* if there are objects $\Delta(i)$ such that:

(st1) there is an epimorphism $\Delta(i)\twoheadrightarrow L(i)$ whose kernel has composition factors $L(j)$ with $j<i$ only;
(st2) there is an epimorphism $P(i)\twoheadrightarrow\Delta(i)$ whose kernel lies in $\mathcal{F}(\Delta(j):j>i)$.

The *costandard objects* $\nabla(i)$ are defined dually, using injective hulls.

By [CPS88], highest weight categories are exactly the module categories of quasi-hereditary algebras. [BB26] also allows partial orders. Since every refinement of the partial order gives an equivalent highest weight structure [BB26, Rem. 2.3], we restrict to total orders throughout.

Definition 2.18 ([DR92]; [BB26, §2.2]). Let Q,T be objects of an abelian category with $V:=\operatorname{Ext}^1(Q,T)$ finite-dimensional. The *universal extension* of Q by T is the extension

$$0\to T\otimes V^\vee\to R\to Q\to 0$$

classified by the canonical element of $\operatorname{Ext}^1(Q,T\otimes V^\vee)\cong V\otimes V^\vee\cong\operatorname{End}(V)$.

The *iterated universal extension* $G=\bigoplus_{s=1}^m G_s$ of a sequence $(T_1,\dots,T_m)$ is defined recursively. For $m=1$ put $G_1=T_1$. For $m>1$, let $G'=\bigoplus_{s<m}G'_s$ be the iterated universal extension of $(T_1,\dots,T_{m-1})$. Then G_s is the universal extension of G'_s by T_m for $s<m$, and $G_m:=T_m$.

A sequence $(E_1,\dots,E_m)$ of objects of $\mathcal{A}$ is *standardizable* if all $\operatorname{Hom}(E_i,E_j)$ and $\operatorname{Ext}^1(E_i,E_j)$ are finite-dimensional, and $\operatorname{rad}(E_i,E_j)=0=\operatorname{Ext}^1(E_i,E_j)$ for $i\geq j$.

Theorem 2.19 (Dlab–Ringel [DR92, Thm. 2]; [BB26, Thm. 2.1]). *Let $(E_1,\dots,E_m)$ be a standardizable sequence in a $\Bbbk$-linear abelian category $\mathcal{A}$, with iterated universal extension G. Then $\widetilde{\mathcal{A}}=\operatorname{mod}\operatorname{End}_{\mathcal{A}}(G)$, with the order $1<\cdots<m$, is a highest weight category with standard objects $\operatorname{Hom}_{\mathcal{A}}(G,E_i)$. Moreover $\mathcal{F}(E_1,\dots,E_m)\subset\mathcal{A}$ and $\mathcal{F}(\Delta(1),\dots,\Delta(m))\subset\widetilde{\mathcal{A}}$ are equivalent.*

Theorem 2.20 ([BB26, Thm. 3.1]). *Let $(E_1,\dots,E_n)$ be a full exceptional sequence in $\mathcal{D}$, let $(F_n,\dots,F_1)$ be its left dual, and let $\mathcal{A}$ be the heart of the t-structure glued along $(E_1,\dots,E_n)$.*

(1) *$\mathcal{A}$ is highest weight with standard objects $\{E_i\}$, costandard objects $\{F_i\}$ and the order induced by the sequence if and only if*

$$\operatorname{Hom}(\textstyle\bigoplus_i E_i,\bigoplus_i E_i[l])=0\quad\text{and}\quad\operatorname{Hom}(\bigoplus_i F_i,\bigoplus_i F_i[l])=0\qquad\text{for all } l<0.$$

(2) *If $\mathcal{D}$ is idempotent complete and dg enhanced and the conditions of (1) hold, then $\mathcal{D}\simeq\mathcal{D}^b(\mathcal{A})$.*

Theorem 2.21 ([BB26, Thm. 3.4]). *Let $\mathcal{D}$ carry a t-structure with heart $\mathcal{A}$, and let $(E_1,\dots,E_n)$ be an exceptional sequence in $\mathcal{D}$ with left dual $(F_n,\dots,F_1)$.*

(1) *If $E_i\in\mathcal{D}^{\leq 0}$ and $F_i\in\mathcal{D}^{\geq 0}$ for all i, then the t-structure of $\mathcal{D}$ restricts to the t-structure on $\mathcal{C}:=\langle E_1,\dots,E_n\rangle$ glued along $(E_1,\dots,E_n)$. Its heart is $\mathcal{A}\cap\mathcal{C}$.*
(2) *If $E_i,F_i\in\mathcal{A}$ for all i, then $\mathcal{A}\cap\mathcal{C}$ is highest weight with standard objects $\{E_i\}$ and costandard objects $\{F_i\}$.*

Theorem 2.22 ([BB26, Thm. 3.8]). *Let $\mathcal{A}$ be an abelian category. Equivalence classes of highest weight structures on $\mathcal{A}$ are in bijection with equivalence classes of full exceptional sequences in $\mathcal{D}^b(\mathcal{A})$ such that both the sequence and its left dual consist of objects of $\mathcal{A}$.*

Example 2.23 ([BB26, Ex. 5.11]). Let $f\colon X \to Y$ be the minimal resolution of an A_1 surface singularity, with exceptional curve E. Let $\mathcal{C}_f = \{F \in \mathcal{D}^b(X) : Rf_*F = 0\}$, and let $\mathcal{A}_f = \mathcal{C}_f \cap \mathrm{Coh}(X)$.

- $\mathcal{A}_f$ is equivalent to $\operatorname{mod}\Bbbk$, with additive generator $\mathcal{O}_E(-1)$.
- $\mathcal{O}_E(-1)$ is 2-spherical. In particular $\mathrm{Ext}^2(\mathcal{O}_E(-1), \mathcal{O}_E(-1)) \cong \Bbbk$, so $\mathcal{D}^b(\mathcal{A}_f) \not\simeq \mathcal{C}_f$.

In the terminology of Section 3, $\mathcal{O}_E(-1)$ is a $\mathbb{P}^1[2]$-object generating $\mathcal{C}_f$. This example already exhibits the phenomenon of Proposition 3.14(1).

Remark 2.24. Kalck's algebra [Kal17, Prop. 5.6] has a full exceptional sequence of modules but is not quasi-hereditary. Bodzenta and Bondal show that its standard t-structure is nevertheless glued along this sequence, and that the dual sequence does not consist of modules [BB26, §4.1]. The condition on the dual sequence in Theorems 2.20 and 2.22 is therefore essential.

2.7. Stratified algebras.

Definition 2.25 ([CPS96, Dla96, AHLU00]). Let $\mathcal{A}$ be Deligne finite, with simple objects $L(1), \dots, L(m)$ indexed by $\{1 < \cdots < m\}$, projective covers $P(i)$ and injective hulls $I(i)$. Define the following objects.

- The *standard object* $\Delta(i) = P(i)/U_i$, where $U_i = \sum_{j>i} \operatorname{tr}_{P(j)} P(i)$. It is the largest quotient of $P(i)$ all of whose composition factors are $L(j)$ with $j \le i$.
- The *proper standard object* $\overline{\Delta}(i)$: the largest quotient of $P(i)$ such that $[\overline{\Delta}(i) : L(i)] = 1$ and all other composition factors are $L(j)$ with $j < i$.
- The *costandard object* $\nabla(i)$: the largest subobject of $I(i)$ all of whose composition factors are $L(j)$ with $j \le i$.
- The *proper costandard object* $\overline{\nabla}(i)$: the largest subobject of $I(i)$ such that $[\overline{\nabla}(i) : L(i)] = 1$ and all other composition factors are $L(j)$ with $j < i$.

We put $B_i := \mathrm{End}_{\mathcal{A}}(\Delta(i))$. The category $\mathcal{A}$ (or an algebra Λ with $\mathcal{A} \simeq \operatorname{mod}\Lambda$) is

- *standardly stratified* if $U_i \in \mathcal{F}(\Delta(j) : j > i)$ for every i, i.e. $P(i)$ has a Δ-filtration with top factor $\Delta(i)$ and remaining factors $\Delta(j)$, $j > i$;
- *properly stratified* if moreover every $P(i)$ lies in $\mathcal{F}(\overline{\Delta})$;
- *quasi-hereditary* if it is standardly stratified and $\Delta(i) = \overline{\Delta}(i)$ for all i.

Remark 2.26. Several inequivalent notions of "stratified" are in use: some allow partial preorders, and the notion of [CPS96] is homological. For total orders, the definition above is equivalent to $\Lambda_\Lambda \in \mathcal{F}(\Delta)$ [Dla96, AHLU00]; Mazorchuk [Maz04] calls this *strongly standardly stratified*. Quasi-hereditary in the above sense is the same as highest weight in the sense of Definition 2.17.

Intuition. The local algebra B_i is the "stratum algebra" at i. Quasi-heredity means $B_i = \Bbbk$ for all i. In the derived picture, $\Delta(i)$ plays the role of a projective generator of the i-th stratum, and $\overline{\nabla}(i)$ the role of its simple object (Lemma 2.27). For $B_i \neq \Bbbk$ these are genuinely different objects.

Lemma 2.27. *Let $\mathcal{A}$ be standardly stratified.*

(1) *Let $X \in \mathcal{F}(\Delta(j) : j > i)$, and let $Y \in \mathcal{A}$ have all composition factors in $\{L(j) : j \le i\}$. Then $\mathrm{Ext}^l_{\mathcal{A}}(X, Y) = 0$ for all $l \ge 0$.*

(2) $\mathrm{Ext}^l(\Delta(j),\Delta(i)) = 0$ *for* $j > i$ *and all* $l \geq 0$, *and* $\mathrm{Ext}^l(\Delta(i),\Delta(i)) = 0$ *for* $l \geq 1$. *The algebra* B_i *is local, and* $\dim B_i = [\Delta(i) : L(i)]$. *In particular* $\mathcal{A}$ *is quasi-hereditary if and only if* $B_i \cong \Bbbk$ *for all* i.

(3) $\mathrm{Ext}^l(\Delta(j),\overline{\nabla}(i)) = 0$ *for* $j > i$ *and all* $l \geq 0$. *Furthermore* $\mathrm{Hom}(\Delta(i),\overline{\nabla}(i)) \cong \Bbbk$ *and* $\mathrm{Ext}^l(\Delta(i),\overline{\nabla}(i)) = 0$ *for* $l \geq 1$, *while* $\mathrm{Hom}(\Delta(j),\overline{\nabla}(i)) = 0$ *for* $j < i$.

(4) *Every* $\Delta(i)$ *has finite projective dimension, and* $K^b(\mathrm{proj}\,\mathcal{A}) = \langle \Delta(1), \ldots, \Delta(m)\rangle$.

Proof. (1) It suffices to treat $X = \Delta(j)$ with $j > i$; we use descending induction on j.

- Since $\Delta(j)$ has top $L(j)$ and Y has no composition factor $L(j)$, we get $\mathrm{Hom}(\Delta(j), Y) = 0$.
- The exact sequence $0 \to U_j \to P(j) \to \Delta(j) \to 0$, together with $\mathrm{Hom}(P(j), Y) = 0$, gives $\mathrm{Ext}^1(\Delta(j), Y) \cong \mathrm{Hom}(U_j, Y)$ and $\mathrm{Ext}^{l+1}(\Delta(j), Y) \cong \mathrm{Ext}^l(U_j, Y)$ for $l \geq 1$.
- Since $U_j \in \mathcal{F}(\Delta(t) : t > j)$, the induction hypothesis gives $\mathrm{Ext}^\bullet(U_j, Y) = 0$. For $j = m$ we have $U_m = 0$.

(2) The first two vanishing statements follow from (1) applied to $Y = \Delta(i)$, using the sequence $0 \to U_i \to P(i) \to \Delta(i) \to 0$ as in (1). Since $\Delta(i)$ has simple top, it is indecomposable, so B_i is local. Every map $P(i) \to \Delta(i)$ vanishes on U_i, because $\mathrm{Hom}(U_i, \Delta(i)) = 0$ by (1). Hence $B_i = \mathrm{Hom}(P(i), \Delta(i))$, whose dimension is $[\Delta(i) : L(i)]$. The last statement follows, since $\Delta(i) = \overline{\Delta}(i)$ if and only if $[\Delta(i) : L(i)] = 1$.

(3) The first statement is (1) with $Y = \overline{\nabla}(i)$.

- The image of a non-zero map $\Delta(j) \to \overline{\nabla}(i)$ is a quotient of $\Delta(j)$ and contains $\mathrm{soc}\,\overline{\nabla}(i) = L(i)$. This is impossible for $j < i$, and forces $\dim \mathrm{Hom}(\Delta(i),\overline{\nabla}(i)) \leq \dim \mathrm{Hom}(P(i),\overline{\nabla}(i)) = [\overline{\nabla}(i) : L(i)] = 1$ for $j = i$.
- The composite $\Delta(i) \twoheadrightarrow L(i) \hookrightarrow \overline{\nabla}(i)$ is non-zero, so $\mathrm{Hom}(\Delta(i),\overline{\nabla}(i)) \cong \Bbbk$.
- The vanishing of $\mathrm{Ext}^l(\Delta(i),\overline{\nabla}(i))$ for $l \geq 1$ follows from $\mathrm{Ext}^\bullet(U_i,\overline{\nabla}(i)) = 0$, which holds by (1).

(4) We use descending induction: $\Delta(m) = P(m)$, and $0 \to U_i \to P(i) \to \Delta(i) \to 0$ with $U_i \in \mathcal{F}(\Delta(j) : j > i)$. This shows that $\mathrm{pd}\,\Delta(i) < \infty$ and that $P(i) \in \langle \Delta(i), \ldots, \Delta(m)\rangle$. □

3. $\mathbb{P}^n[k]$-like objects and the hearts of the categories they generate

3.1. Definitions and examples.

Definition 3.1 ([Kru18], [HK19, Def. 2.1], [Bonf26, Def. 2.1])**.** Let $n \geq 0$ and $k \in \mathbb{Z}$. An object P of $\mathcal{D}$ is $\mathbb{P}^n[k]$-*like* if there is an isomorphism of graded algebras

$$\mathrm{End}^\bullet(P) \cong \Bbbk[t]/(t^{n+1}), \qquad \deg t = k.$$

It is a $\mathbb{P}^n[k]$-*object* if moreover $\mathrm{Hom}^\bullet(P, X)$ is finite-dimensional for all X and P is nk-Calabi–Yau, i.e. there are isomorphisms $\mathrm{Hom}(P, X) \cong \mathrm{Hom}(X, P[nk])^\vee$, functorial in $X \in \mathcal{D}$.

As in [Bonf26] we allow arbitrary $k \in \mathbb{Z}$. For $k = 0$ the condition says that $\mathrm{End}(P) \cong \Bbbk[t]/(t^{n+1})$ and $\mathrm{Hom}(P, P[l]) = 0$ for $l \neq 0$.

Example 3.2.

(1) The $\mathbb{P}^0[k]$-like objects are exactly the exceptional objects.

(2) $\mathbb{P}^1[k]$-like objects are the k-*spherelike* objects. $\mathbb{P}^1[k]$-objects are the k-spherical objects of Seidel and Thomas [ST01]; in the terminology of [HJY13] they are w-*spherical* with $w = k$.

(3) $\mathbb{P}^n[2]$-objects are the $\mathbb{P}^n$-objects of Huybrechts and Thomas [HT06]. For example, the structure sheaf of a compact hyperkähler manifold of dimension $2n$ is one.

(4) Krug [Kru18] introduced the notation $\mathbb{P}^n[k]$ and gave examples of varieties whose structure sheaves are $\mathbb{P}^n[k]$-objects; see also [HK19].
(5) ($k = 0$) If $B = \Bbbk[t]/(t^{n+1})$, then B is a $\mathbb{P}^n[0]$-like object of $\mathcal{D}^b(\operatorname{mod} B)$. More generally, a module M with $\operatorname{End}(M) \cong \Bbbk[t]/(t^{n+1})$ and $\operatorname{Ext}^{>0}(M, M) = 0$ is $\mathbb{P}^n[0]$-like. Standard modules of stratified algebras provide examples (Theorem 6.3).

Remark 3.3 ([HK19, Rem. 2.2], [Bonf26, Lemma 2.5]). A $\mathbb{P}^n[k]$-like object P is already a $\mathbb{P}^n[k]$-object of the subcategory $\langle P\rangle$; in fact, for $\mathcal{D}$ algebraic, $\langle P\rangle$ is nk-Calabi–Yau. Hence for questions concerning $\langle P\rangle$ alone, the Calabi–Yau condition is irrelevant. In sequences of such objects the Calabi–Yau condition refers to the ambient category and is not preserved by mutations, so we work with "$\mathbb{P}^n[k]$-like" throughout.

Intuition. For an exceptional object E one has $\langle E\rangle \simeq \mathcal{D}^b(\Bbbk)$, and E is at the same time the simple object, the projective object and the injective object of the heart $\operatorname{mod}\Bbbk$. A $\mathbb{P}^n[k]$-like object is a "thickening" with self-extensions in degrees $k, 2k, \dots, nk$, and these three roles separate:

- For $k \geq 1$, P has no negative self-extensions and $\operatorname{End}(P) = \Bbbk$. It is *simple-minded* and determines a t-structure.
- For $k \leq 0$, P has no positive self-extensions, so it is *silting* and determines a co-t-structure. For $k = 0$ it behaves like the free module over the local algebra $\Bbbk[t]/(t^{n+1})$.

This is the dichotomy of Theorems 3.4 and 3.6; the standard t-structure and the standard weight structure on $\langle P\rangle$ are obtained in exactly this way in [Bonf26, §3.2].

3.2. **Known classification results.**

Theorem 3.4 (Holm–Jørgensen–Yang [HJY13])**.** *Let $w \in \mathbb{Z}$, and let $\mathcal{T}$ be the $\Bbbk$-linear algebraic triangulated category generated by a w-spherical object.*

- *For $w \leq 0$, $\mathcal{T}$ has no non-trivial t-structures, but it has one family of non-trivial co-t-structures.*
- *For $w \geq 1$, $\mathcal{T}$ has no non-trivial co-t-structures, but it has one family of non-trivial t-structures.*

For $w = 0$, $\mathcal{T}$ is the compact derived category of the dual numbers.

Here "non-trivial" excludes $(\mathcal{T}, 0)$ and $(0, \mathcal{T})$, and a "family" consists of the shifts of a single structure.

Theorem 3.5 (Hochenegger–Krug [HK19, Lemma 2.8]; Bonfert [Bonf26, Props. 2.3 and 2.4])**.** *For all $n \geq 0$ and $k \in \mathbb{Z}$, the graded algebra $A_{n,k} := \Bbbk[x]/(x^{n+1})$ with $\deg x = k$ is intrinsically formal. Consequently, let $\mathcal{D}$ be an idempotent complete algebraic triangulated category and $P \in \mathcal{D}$ a $\mathbb{P}^n[k]$-like object. Then $\langle P\rangle \simeq \operatorname{perf}(A_{n,k})$, where $A_{n,k}$ is regarded as a dg algebra with zero differential, and P corresponds to $A_{n,k}$.*

Theorem 3.6 (Bonfert [Bonf26, Thm. 3.3])**.** *Let $\mathcal{T}$ be the triangulated category generated by a $\mathbb{P}^n[k]$-object P.*

(1) *If $n = 0$, every non-trivial t-structure on $\mathcal{T}$ is a shift of the standard t-structure, and every non-trivial weight structure on $\mathcal{T}$ is a shift of the standard weight structure.*
(2) *If $n > 0$ and $k > 0$, every non-trivial t-structure on $\mathcal{T}$ is a shift of the standard t-structure, whose heart is* $\operatorname{Filt}(P)$*. There is no non-trivial weight structure on $\mathcal{T}$.*
(3) *If $n > 0$ and $k \leq 0$, every non-trivial weight structure on $\mathcal{T}$ is a shift of the standard weight structure, whose co-heart is* $\operatorname{add} P$*. There is no non-trivial t-structure on $\mathcal{T}$.*

For $n = 1$ this recovers Theorem 3.4. By Theorem 3.5 and the preceding remark, Theorem 3.6 applies to $\langle P\rangle$ for every $\mathbb{P}^n[k]$-like object P of an idempotent complete algebraic triangulated category.

Remark 3.7 (Koszul dual models [Bonf26, §4]). Bonfert also computes the dg Koszul dual $A^{!,\mathrm{dg}}_{n,k} = \mathbf{R}\mathrm{Hom}_{A_{n,k}}(\Bbbk,\Bbbk)$ [Bonf26, Thm. 4.9, Props. 4.11 and 4.12]:

- for $n = 1$ and $k \neq 1$ it is $\Bbbk[s]$ with $\deg s = 1-k$;
- for $n = k = 1$ it is $\Bbbk[[s]]$ with $\deg s = 0$;
- for $n > 1$ it is a non-formal dg algebra with cohomology $\Bbbk[s,t]/(s^2)$, where $\deg s = 1-k$ and $\deg t = 2-(n+1)k$.

For $k = 0$ this is the Ext-algebra $\mathrm{Ext}^\bullet_B(\Bbbk,\Bbbk)$ of the simple module over $B = \Bbbk[t]/(t^{n+1})$. It is the graded endomorphism algebra of the "simple-like" objects $\overline{\nabla}$ that will be paired with $\mathbb{P}^n[0]$-like objects in Sections 4 and 6.

3.3. **The case $k = 0$.**

Lemma 3.8. *Let B be a finite-dimensional local algebra, and let $Y \in K^b(\mathrm{proj}\, B)$. Then $\chi(Y) := \sum_l (-1)^l \dim H^l(Y)$ is divisible by $\dim B$.*

Proof. Finitely generated projective modules over a local algebra are free. So Y is isomorphic to a bounded complex with terms B^{r_l}, and $\chi(Y) = \sum_l (-1)^l r_l \dim B$. □

Proposition 3.9 (cf. [Bonf26, Prop. 2.4, §3.2, Cor. 3.10])**.** *Let $\mathcal{D}$ be algebraic, let $P \in \mathcal{D}$ be $\mathbb{P}^n[0]$-like with $n \geq 1$, and put $B = \Bbbk[t]/(t^{n+1})$.*

1. *P is a tilting object of $\langle P\rangle$, and $\langle P\rangle \simeq K^b(\mathrm{proj}\, B)$ with $P \mapsto B$.*
2. *$\langle P\rangle$ carries the bounded co-t-structure with co-heart $\mathrm{add}\, P$.*
3. *There is no $X \in \langle P\rangle$ with $\mathrm{Hom}^\bullet(P,X) \cong \Bbbk$ concentrated in one degree. In particular, the standard t-structure of $\mathcal{D}^b(\mathrm{mod}\, B)$ does not restrict to $K^b(\mathrm{proj}\, B)$.*

Proof. (1) follows from Theorem 3.5 (or directly from Theorem 2.10), and (2) from Theorem 2.8.

(3) Under the equivalence of (1), $\mathrm{Hom}^l(P,X) \cong H^l(Y)$, where Y is the perfect complex corresponding to X. By Lemma 3.8, $\chi(Y)$ is divisible by $n+1 \geq 2$, so it cannot equal ± 1.

For the last assertion, consider the complex $Y = (B \xrightarrow{t} B)$ in degrees $-1, 0$. We have $H^0(Y) \cong \Bbbk$. If the standard t-structure restricted to $K^b(\mathrm{proj}\, B)$, then $H^0(Y) = \tau^{\geq 0}\tau^{\leq 0}Y$ would be perfect, contradicting what we just proved. □

Remark 3.10. Parts (1) and (2) of Proposition 3.9 are contained in [Bonf26, Prop. 2.4 and §3.2]. Part (3) is a weak form of [Bonf26, Cor. 3.10], proved here by a different (Euler characteristic) argument, which is reused in the proof of Theorem 6.3. By Theorem 3.6(3) (for $n = 1$ already by Theorem 3.4), $\langle P\rangle$ has no non-trivial t-structure at all. The object "missing" from $\langle P\rangle$ is the simple B-module, whose role is played in Sections 4 and 6 by the proper costandard objects. Note that the simple module is Koszul dual to B: its graded endomorphism algebra is $\mathrm{Ext}^\bullet_B(\Bbbk,\Bbbk)$, which is infinite-dimensional (Remark 3.7).

3.4. **The case $k \geq 1$.** The following lemma describes the aisles of a stratum in terms of two "test objects". It is the local ingredient in the description of glued aisles (Proposition 4.5).

Lemma 3.11. *Let $\mathcal{T}$ carry a bounded t-structure whose heart $\mathcal{H}$ is a length category with a unique simple object S (up to isomorphism). Let $Q \in \mathcal{H}$ satisfy $\mathrm{Hom}(Q,M) \neq 0$ for every non-zero $M \in \mathcal{H}$. Then*

$$\mathcal{T}^{\leq 0} = \{X : \mathrm{Hom}(X, S[l]) = 0 \ \forall l < 0\}, \qquad \mathcal{T}^{\geq 0} = \{X : \mathrm{Hom}(Q, X[l]) = 0 \ \forall l < 0\}.$$

Moreover $\mathcal{T}^{\leq 0} = \mathrm{Filt}\{S[l] : l \geq 0\}$ *and* $\mathcal{T}^{\geq 0} = \mathrm{Filt}\{S[l] : l \leq 0\}$. *The hypothesis on* Q *holds for* $Q = S$, *and for a projective cover of* S *if one exists.*

Proof. **Description of $\mathcal{T}^{\geq 0}$.** If $X \in \mathcal{T}^{\geq 0}$ and $l < 0$, then $X[l] \in \mathcal{T}^{\geq 1}$, so $\mathrm{Hom}(Q, X[l]) = 0$. Conversely, suppose $X \notin \mathcal{T}^{\geq 0}$. By boundedness there is a minimal $d < 0$ with $H^d(X) \neq 0$, and we have a triangle $H^d(X)[-d] \to X \to \tau^{\geq d+1}X \to$. Shifting by d and applying $\mathrm{Hom}(Q, -)$ gives an exact sequence

$$\mathrm{Hom}(Q, (\tau^{\geq d+1}X)[d-1]) \to \mathrm{Hom}(Q, H^d(X)) \to \mathrm{Hom}(Q, X[d]).$$

The first term vanishes because $(\tau^{\geq d+1}X)[d-1] \in \mathcal{T}^{\geq 2}$. The middle term is non-zero by hypothesis. Hence $\mathrm{Hom}(Q, X[d]) \neq 0$ with $d < 0$.

Description of $\mathcal{T}^{\leq 0}$. If $X \in \mathcal{T}^{\leq 0}$ and $l < 0$, then $S[l] \in \mathcal{T}^{\geq 1}$, so $\mathrm{Hom}(X, S[l]) = 0$. Conversely, suppose $X \notin \mathcal{T}^{\leq 0}$, and let $d > 0$ be maximal with $H^d(X) \neq 0$. Since $X \in \mathcal{T}^{\leq d}$, for every $M \in \mathcal{H}$ we have

$$\mathrm{Hom}(X, M[-d]) \cong \mathrm{Hom}(\tau^{\geq d}X, M[-d]) \cong \mathrm{Hom}(H^d(X), M).$$

Taking $M = S$ and using an epimorphism $H^d(X) \twoheadrightarrow S$, we obtain $\mathrm{Hom}(X, S[-d]) \neq 0$ with $-d < 0$.

The remaining statements. The descriptions via Filt hold because $\mathcal{H} = \mathrm{Filt}(S)$ and the t-structure is bounded. Every non-zero $M \in \mathcal{H}$ has a simple subobject (isomorphic to S) and a simple quotient; the projection from a projective cover of S onto the simple quotient lifts to M. This gives the last assertion. □

Proposition 3.12 (cf. [Bonf26, §3.2])**.** *Let* P *be* $\mathbb{P}^n[k]$*-like with* $k \geq 1$. *Then* $\{P\}$ *is a simple-minded collection in* $\langle P \rangle$. *Consequently* $\langle P \rangle$ *carries a bounded t-structure* t_P *with heart* $\mathcal{H}_P := \mathrm{Filt}(P)$, *a length category with unique simple object* P. *The aisles of* t_P *are described by Lemma 3.11 with* $Q = S = P$.

Proof. Condition (SM1) holds because $\mathrm{End}^\bullet(P)$ is concentrated in degrees $0, k, \ldots, nk \geq 0$. Condition (SM2) holds because $\mathrm{End}(P) = \Bbbk$ (as $k \neq 0$), and (SM3) is clear. Now apply Theorem 2.6, and Lemma 3.11 with $Q = S = P$. □

Remark 3.13. The t-structure t_P is the standard t-structure of the proper positive dg algebra $A_{n,k}$ [KN13, Thm. 7.1], [Bonf26, §3.2]. By Theorem 3.6(2), every non-trivial t-structure on $\langle P \rangle$ is a shift of t_P. Hence, for strata generated by $\mathbb{P}^n[k]$-like objects with $k \geq 1$, the gluing data used in Sections 4 and 5 are determined up to shifts.

Proposition 3.14 (cf. [Bonf26, Lemmas 3.11 and 4.15, Cor. 4.16])**.** *Let* P *be* $\mathbb{P}^n[k]$*-like with* $k \geq 1$ *and* $n \geq 1$, *and let* $\mathcal{H}_P$ *be as in Proposition 3.12.*

(1) *If* $k \geq 2$, *then* $\mathcal{H}_P = \mathrm{add}\, P \simeq \mathrm{mod}\, \Bbbk$.
(2) *If* $k = n = 1$, *then* $\mathcal{H}_P$ *is hereditary and has no non-zero projective objects. If* $\langle P \rangle$ *is algebraic, then* $\mathrm{real}\colon \mathcal{D}^b(\mathcal{H}_P) \to \langle P \rangle$ *is an equivalence.*
(3) *Let* $k = 1$ *and* $n \geq 2$. *Let* t *span* $\mathrm{Hom}(P, P[1])$, *and define* Q *by the triangle*

$$P \xrightarrow{a} Q \xrightarrow{b} P \xrightarrow{t} P[1].$$

Then:

- $Q \in \mathcal{H}_P$ *is a projective cover of* P *in* $\mathcal{H}_P$, $\mathrm{End}(Q) \cong \Bbbk[x]/(x^2)$, *and* $\mathcal{H}_P \simeq \mathrm{mod}\, \Bbbk[x]/(x^2)$;
- $\mathrm{Hom}(Q, P) \cong \Bbbk$, *and for* $l \neq 0$ *we have* $\mathrm{Hom}(Q, P[l]) \cong \Bbbk$ *if* $l = n$ *and* $\mathrm{Hom}(Q, P[l]) = 0$ *otherwise;*
- $\mathrm{Hom}(Q, Q[1]) = 0$.

(4) *In cases* (1) *and* (3), *the comparison maps* $\mathrm{Ext}^\bullet_{\mathcal{H}_P}(P,P) \to \mathrm{End}^\bullet(P)$ *are not bijective. In particular, no realisation functor* $\mathcal{D}^b(\mathcal{H}_P) \to \langle P\rangle$ *is an equivalence.*

Proof. Recall that $\mathrm{Ext}^1_{\mathcal{H}_P}(X,Y) = \mathrm{Hom}(X, Y[1])$ for $X, Y \in \mathcal{H}_P$ (Proposition 2.2).

(1) We have $\mathrm{Ext}^1_{\mathcal{H}_P}(P,P) = \mathrm{End}^1(P) = 0$ since $k \geq 2$. Hence every object of $\mathrm{Filt}(P)$ is isomorphic to some $P^{\oplus r}$, and $\mathrm{Hom}(P,-)$ gives $\mathcal{H}_P \simeq \operatorname{mod} \Bbbk$.

(2) We have $\mathrm{Ext}^1_{\mathcal{H}_P}(P,P) \cong \Bbbk$. By Proposition 2.2, $\mathrm{Ext}^2_{\mathcal{H}_P}(P,P)$ embeds into $\mathrm{End}^2(P)$, which is 0 for $(k,n) = (1,1)$.

- By dévissage (long exact sequences in both variables), $\mathrm{Ext}^2_{\mathcal{H}_P}$ vanishes identically. Since Yoneda extensions of length ≥ 2 factor through Yoneda products with Ext^2, all $\mathrm{Ext}^{\geq 2}_{\mathcal{H}_P}$ vanish, so $\mathcal{H}_P$ is hereditary.
- Consequently $\chi(X,Y) := \dim \mathrm{Hom}(X,Y) - \dim \mathrm{Ext}^1(X,Y)$ is additive in both variables. It is therefore given by $\chi(X,Y) = \ell(X)\ell(Y)\chi(P,P) = 0$, where ℓ denotes length.
- If R is projective, then $\dim \mathrm{End}(R) = \chi(R,R) = 0$, so $R = 0$.
- The comparison maps $\mathrm{Ext}^l_{\mathcal{H}_P}(P,P) \to \mathrm{End}^l(P)$ are bijective for $l \leq 1$ by Proposition 2.2, and both sides vanish for $l \geq 2$. By dévissage and the five lemma, $\mathrm{Ext}^l_{\mathcal{H}_P}(X,Y) \to \mathrm{Hom}(X, Y[l])$ is bijective for all $X, Y \in \mathcal{H}_P$ and all l. Thus real is fully faithful, and it is an equivalence by Theorem 2.3.

(3) Since $\mathcal{H}_P$ is closed under extensions, $Q \in \mathcal{H}_P$, and $0 \to P \to Q \to P \to 0$ is a non-split exact sequence in $\mathcal{H}_P$. Let $W \in \langle P\rangle$. Applying $\mathrm{Hom}(-,W)$ to the triangle gives an exact sequence

$$\mathrm{Hom}(P[1],W) \xrightarrow{t^*} \mathrm{Hom}(P,W) \xrightarrow{b^*} \mathrm{Hom}(Q,W) \xrightarrow{a^*} \mathrm{Hom}(P,W) \xrightarrow{t[-1]^*} \mathrm{Hom}(P[-1],W).$$

For $W = P[l]$, the outer maps are, up to sign, multiplication by t, namely $\mathrm{End}^{l-1}(P) \to \mathrm{End}^l(P)$ and $\mathrm{End}^l(P) \to \mathrm{End}^{l+1}(P)$. Hence there is a short exact sequence

$$0 \to \operatorname{coker}\left(\mathrm{End}^{l-1}(P) \xrightarrow{t} \mathrm{End}^l(P)\right) \to \mathrm{Hom}(Q, P[l]) \to \ker\left(\mathrm{End}^l(P) \xrightarrow{t} \mathrm{End}^{l+1}(P)\right) \to 0.$$

Since $\deg t = 1$ and $t^n \neq 0 = t^{n+1}$:

- the cokernel is $\Bbbk$ for $l = 0$ and 0 otherwise;
- the kernel is $\Bbbk$ (spanned by t^n) for $l = n$ and 0 otherwise.

This proves the second item of (3); note that $0 \neq n$.

Since $n \geq 2$, we have $\mathrm{Hom}(Q, P[1]) = 0$, i.e. $\mathrm{Ext}^1_{\mathcal{H}_P}(Q,P) = 0$. As P is the only simple object, dévissage gives $\mathrm{Ext}^1_{\mathcal{H}_P}(Q,-) = 0$, so Q is projective. Since $\mathrm{Hom}(Q,P) \cong \Bbbk$, the top of Q is P, and Q is a projective cover of P.

Applying $\mathrm{Hom}(Q,-)$ to the triangle gives an exact sequence

$$\mathrm{Hom}(Q, P[-1]) \to \mathrm{Hom}(Q,P) \to \mathrm{End}(Q) \to \mathrm{Hom}(Q,P) \to \mathrm{Hom}(Q, P[1]).$$

Its outer terms vanish, so $\dim \mathrm{End}(Q) = 2$. The algebra $\mathrm{End}(Q)$ is local because Q has simple top; hence $\mathrm{End}(Q) \cong \Bbbk[x]/(x^2)$. A Hom-finite length category with a projective generator Q is equivalent to $\operatorname{mod} \mathrm{End}(Q)$ via $\mathrm{Hom}(Q,-)$, so $\mathcal{H}_P \simeq \operatorname{mod} \Bbbk[x]/(x^2)$. Finally, $\mathrm{Hom}(Q, Q[1]) = \mathrm{Ext}^1_{\mathcal{H}_P}(Q,Q) = 0$.

(4) In case (1), $\mathrm{Ext}^{nk}_{\mathcal{H}_P}(P,P) = 0$ while $\mathrm{End}^{nk}(P) \neq 0$. In case (3), $\mathrm{Ext}^{n+1}_{\Bbbk[x]/(x^2)}(\Bbbk,\Bbbk) \cong \Bbbk$, because the minimal projective resolution of $\Bbbk$ is periodic, while $\mathrm{End}^{n+1}(P) = 0$. A realisation functor induces the comparison maps on extension groups, so it cannot be fully faithful. □

Remark 3.15. Parts (1)–(3) overlap with [Bonf26, Lemma 3.11], where the indecomposable objects of $\mathcal{H}_P$ are listed and the spaces $\mathrm{Hom}(X_2, P[i])$ are computed; our Q is Bonfert's X_2.

They also overlap with [Bonf26, Lemma 4.15], which shows that $\mathcal{H}_P$ has an injective cogenerator unless $n = k = 1$. The realisation statement in (2) corresponds to Bonfert's Koszul-dual model [Bonf26, Cor. 4.16(2)], which identifies $\langle P\rangle$ with the thick subcategory of $\mathcal{D}(\Bbbk[[t]])$ generated by the simple module. What is new in Proposition 3.14 is only the explicit identification $\mathcal{H}_P \simeq \operatorname{mod}\Bbbk[x]/(x^2)$ in (3) and the reformulation of these facts in terms of the conditions ($\star$) and ($\star\star$) used for gluing. We have given the proofs in the form needed below.

Remark 3.16.

(1) For $n = 1$, parts (1) and (2) are consistent with Theorem 3.4. For $w = 1$, $\langle P\rangle$ is the derived category of finite-dimensional nilpotent $\Bbbk[x]$-modules; cf. [Bonf26, Lemma 3.11(3), Cor. 4.16(2)]. For $w = 2$, part (1) is exactly the phenomenon of Example 2.23.

(2) The simple module $\Bbbk$ generates $\mathcal{D}^b(\operatorname{mod}\Bbbk[x]/(x^2))$, and $\operatorname{End}^\bullet(\Bbbk) \cong \Bbbk[s]$ with $\deg s = 1$. So $\Bbbk$ is a "$\mathbb{P}^\infty[1]$-like" object (Remark 3.7 with $n = 1$, $k = 0$). A $\mathbb{P}^n[1]$-like object with $n \geq 2$ has the same heart, but its endomorphism algebra is truncated at $t^{n+1} = 0$. Meanwhile, the regular module $\Bbbk[x]/(x^2)$ is $\mathbb{P}^1[0]$-like. Thus the cases $k = 0$ and $k = 1$ are Koszul dual to each other, which is the mechanism behind Theorem 3.4.

(3) The non-vanishing $\operatorname{Hom}(Q, P[n]) \neq 0$ in (3) and $\operatorname{End}^k(P) \neq 0$ in (1) are exactly the obstructions to the condition ($\star\star$) of Theorem 4.8. The vanishing of $\operatorname{Hom}(Q, P[2])$, i.e. $n \neq 2$, and of $\operatorname{End}^2(P)$, i.e. $k \neq 2$, is condition ($\star$).

4. Gluing strata with local hearts

4.1. Setting.

Setting 4.1. Let $\mathcal{D}$ be a triangulated category (Hom-finite and idempotent complete, as always) with an admissible filtration $0 = \mathcal{D}_0 \subset \mathcal{D}_1 \subset \cdots \subset \mathcal{D}_m = \mathcal{D}$ (Definition 2.14) with strata $\mathcal{T}_s = \mathcal{D}_s/\mathcal{D}_{s-1}$. Assume:

- each $\mathcal{T}_s$ carries a bounded t-structure whose heart $\mathcal{H}_s$ is a length category with a unique simple object S_s up to isomorphism;
- $\mathcal{D}$ carries the glued t-structure, with heart $\mathcal{A}$; we write $\mathcal{A}_s = \mathcal{A} \cap \mathcal{D}_s$ for the heart of $\mathcal{D}_s$.

We say that we are in the *projective case* if each S_s has a projective cover Π_s in $\mathcal{H}_s$. In this case we put $B_s := \operatorname{End}_{\mathcal{H}_s}(\Pi_s)$; this is a finite-dimensional local algebra, and $\mathcal{H}_s \simeq \operatorname{mod} B_s$.

Notation 4.2. In Setting 4.1 we define:

- $L_s := j_{s!*}S_s \in \mathcal{A}_s \subset \mathcal{A}$; by Proposition 2.13(3) and induction, $L_1, \ldots, L_m$ are the simple objects of $\mathcal{A}$;
- $\overline{\nabla}_s := j_{s*}S_s \in \mathcal{D}_s \subset \mathcal{D}$.

A family $Q = (Q_s)_s$ of objects $Q_s \in \mathcal{H}_s$ with $\operatorname{Hom}(Q_s, M) \neq 0$ for all non-zero $M \in \mathcal{H}_s$ is called a family of *test objects*; for instance $Q_s = S_s$, or $Q_s = \Pi_s$ in the projective case (Lemma 3.11). For such a family we put $\Delta_s^Q := j_{s!}Q_s$. In the projective case we write

$$\Delta_s := \Delta_s^\Pi = j_{s!}\Pi_s.$$

Example 4.3.

(1) *Exceptional sequences.* If $\mathcal{T}_s \simeq \mathcal{D}^b(\Bbbk)$ with the standard t-structure, then $\Pi_s = S_s = \Bbbk$, and $\Delta_s = E_s$, $\overline{\nabla}_s = F_s$ are a dual pair of exceptional sequences.

(2) *Derived categories of local algebras.* If $\mathcal{T}_s \simeq \mathcal{D}^b(\operatorname{mod} B_s)$ with the standard t-structure, then $\Pi_s = B_s$ and $S_s = \Bbbk$.

(3) $\mathbb{P}^n[k]$-*like objects.* If $\mathcal{T}_s = \langle P_s \rangle$ with P_s $\mathbb{P}^{n_s}[k_s]$-like, $k_s \geq 1$, and t_{P_s} as in Proposition 3.12, then $S_s = P_s$. By Proposition 3.14, we are in the projective case if and only if $n_s = 0$, or $k_s \geq 2$, or ($k_s = 1$ and $n_s \geq 2$).

Intuition. Δ_s is the projective object of the s-th stratum extended by zero, and $\overline{\nabla}_s$ is the simple object of the s-th stratum extended by j_{s*}. In the exceptional case the projective and the simple object of the stratum coincide, so Δ_s and $\overline{\nabla}_s$ belong to a dual pair of exceptional sequences. In general they are of a different nature: one is "projective-like", the other "simple-like". This asymmetric pairing is what makes the Hom-duality of Lemma 4.4 hold. It replaces the Hom-duality (1) of exceptional sequences, which is not available, since mutation does not exchange projectives and simples.

4.2. **Hom-duality and the glued aisles.**

Lemma 4.4 (Hom-duality)**.** *In Setting 4.1, with a family of test objects Q:*

(1) $\mathrm{Hom}^\bullet(\Delta_s^Q, \overline{\nabla}_t) = 0$ *for* $s \neq t$;
(2) $\mathrm{Hom}^\bullet(\Delta_s^Q, \overline{\nabla}_s) \cong \mathrm{Hom}^\bullet_{\mathcal{T}_s}(Q_s, S_s)$. *This is concentrated in degrees* ≥ 0, *and its degree* 0 *part is non-zero.*

In the projective case, $\mathrm{Hom}^0(\Delta_s, \overline{\nabla}_s) \cong \Bbbk$.

Proof. Recall that $j_{s!}$ identifies $\mathcal{T}_s$ with ${}^\perp\mathcal{D}_{s-1} \cap \mathcal{D}_s$, and j_{s*} identifies it with $\mathcal{D}_{s-1}^\perp \cap \mathcal{D}_s$.

- If $s < t$, then $\Delta_s^Q \in \mathcal{D}_s \subset \mathcal{D}_{t-1}$ while $\overline{\nabla}_t \in \mathcal{D}_{t-1}^\perp$.
- If $s > t$, then $\overline{\nabla}_t \in \mathcal{D}_t \subset \mathcal{D}_{s-1}$ while $\Delta_s^Q \in {}^\perp\mathcal{D}_{s-1}$.
- If $s = t$, adjunction and $j_s^* j_{s!} \cong \mathrm{id}$ give $\mathrm{Hom}(j_{s!}Q_s, j_{s*}S_s[l]) \cong \mathrm{Hom}(Q_s, S_s[l])$. This vanishes for $l < 0$ since $Q_s, S_s \in \mathcal{H}_s$, and it is non-zero for $l = 0$ by the definition of test objects.

For a projective cover Π_s we have $\dim \mathrm{Hom}(\Pi_s, S_s) = [S_s : S_s] = 1$. □

Proposition 4.5 (Glued aisles; cf. Theorem 2.16)**.** *In Setting 4.1, with a family of test objects Q,*

$$\mathcal{D}^{\leq 0} = \{X \in \mathcal{D} : \mathrm{Hom}(X, \overline{\nabla}_s[l]) = 0 \ \forall s, \ \forall l < 0\},$$
$$\mathcal{D}^{\geq 0} = \{X \in \mathcal{D} : \mathrm{Hom}(\Delta_s^Q, X[l]) = 0 \ \forall s, \ \forall l < 0\}.$$

Proof. We use induction on m; the case $m = 1$ is Lemma 3.11. For $m \geq 2$ write $i = i_m$ and $j = j_m$. By Theorem 2.12, $X \in \mathcal{D}^{\leq 0}$ if and only if $i^*X \in \mathcal{D}_{m-1}^{\leq 0}$ and $j^*X \in \mathcal{T}_m^{\leq 0}$.

- By induction, $i^*X \in \mathcal{D}_{m-1}^{\leq 0}$ if and only if $\mathrm{Hom}(i^*X, \overline{\nabla}_s[l]) = 0$ for $s < m$ and $l < 0$. By adjunction this is $\mathrm{Hom}(X, i_*\overline{\nabla}_s[l])$, and the objects $\overline{\nabla}_s$ ($s < m$) of $\mathcal{D}_{m-1}$ and of $\mathcal{D}$ agree.
- By Lemma 3.11, $j^*X \in \mathcal{T}_m^{\leq 0}$ if and only if $\mathrm{Hom}(j^*X, S_m[l]) = \mathrm{Hom}(X, \overline{\nabla}_m[l])$ vanishes for $l < 0$.

The description of $\mathcal{D}^{\geq 0}$ is proved in the same way, using $i^!$, the adjunction $\mathrm{Hom}(i_*\Delta_s^Q, X) = \mathrm{Hom}(\Delta_s^Q, i^!X)$, and $\mathrm{Hom}(Q_m, j^*X[l]) = \mathrm{Hom}(\Delta_m^Q, X[l])$. This is the argument of [BB26, Lemma 2.4], with $\mathrm{Hom}^\bullet(-, F)^\vee$ and $\mathrm{Hom}^\bullet(E, -)$ replaced by the two different test functors of Lemma 3.11. □

Proposition 4.6 (Criterion; cf. Theorem 2.20(1))**.** *In Setting 4.1, with a family of test objects Q, the following are equivalent:*

(a) $\Delta_s^Q \in \mathcal{A}$ *and* $\overline{\nabla}_s \in \mathcal{A}$ *for all* s;
(b) $\mathrm{Hom}(\Delta_s^Q, \Delta_t^Q[l]) = 0$ *and* $\mathrm{Hom}(\overline{\nabla}_s, \overline{\nabla}_t[l]) = 0$ *for all* s, t *and all* $l < 0$.

If these conditions hold, then $j_{s*}M \in \mathcal{A}$ and $j_{s!}N \in \mathcal{A}$ for all $M \in \mathcal{H}_s$ and all $N \in \mathrm{Filt}(Q_s) \subset \mathcal{H}_s$. In the projective case, if $j_{s!}S_s \in \mathcal{A}$, then also $\Delta_s \in \mathcal{A}$.

Proof. (a)⇒(b) holds because $\mathcal{A}$ is a heart.

(b)⇒(a): By Proposition 4.5, $\Delta_s^Q \in \mathcal{D}^{\geq 0}$ is the first condition in (b), and $\Delta_s^Q \in \mathcal{D}^{\leq 0}$ follows from Lemma 4.4. The argument for $\overline{\nabla}_s$ is dual.

The remaining assertions hold because $\mathcal{H}_s = \mathrm{Filt}(S_s)$, $\Pi_s \in \mathrm{Filt}(S_s)$, and j_{s*} and $j_{s!}$ are triangulated functors, while $\mathcal{A}$ is closed under extensions. □

Proposition 4.7 (Recognising a glued t-structure; cf. Theorem 2.21(1))**.** *Let $\mathcal{D}$ be as in Setting 4.1, and suppose $\mathcal{D}$ carries a second t-structure $(\mathcal{D}_\natural^{\leq 0}, \mathcal{D}_\natural^{\geq 0})$. If, for some family of test objects, $\Delta_s^Q \in \mathcal{D}_\natural^{\leq 0}$ and $\overline{\nabla}_s \in \mathcal{D}_\natural^{\geq 0}$ for all s, then $(\mathcal{D}_\natural^{\leq 0}, \mathcal{D}_\natural^{\geq 0})$ coincides with the glued t-structure.*

Proof. Let $X \in \mathcal{D}_\natural^{\leq 0}$ and $l < 0$. Then $\overline{\nabla}_s[l] \in \mathcal{D}_\natural^{\geq 1}$, so $\mathrm{Hom}(X, \overline{\nabla}_s[l]) = 0$. By Proposition 4.5 this gives $\mathcal{D}_\natural^{\leq 0} \subset \mathcal{D}^{\leq 0}$. Dually, $\mathcal{D}_\natural^{\geq 0} \subset \mathcal{D}^{\geq 0}$. Two t-structures with $\mathcal{D}_\natural^{\leq 0} \subset \mathcal{D}^{\leq 0}$ and $\mathcal{D}_\natural^{\geq 0} \subset \mathcal{D}^{\geq 0}$ coincide. Indeed, the second inclusion gives $\mathcal{D}^{\leq 0} = {}^\perp(\mathcal{D}^{\geq 1}) \subset {}^\perp(\mathcal{D}_\natural^{\geq 1}) = \mathcal{D}_\natural^{\leq 0}$, so the aisles agree, and hence so do the co-aisles. □

4.3. **The main theorem.** In the projective case we consider the following two conditions:

$$\mathrm{Hom}_{\mathcal{T}_s}(\Pi_s, S_s[2]) = 0 \qquad \text{for } 1 \leq s \leq m; \tag{$\star$}$$

$$\mathrm{Hom}_{\mathcal{T}_s}(\Pi_s, S_s[l]) = 0 \qquad \text{for } 1 \leq s \leq m \text{ and all } l \geq 1. \tag{$\star\star$}$$

Since $\mathrm{Hom}(\Pi_s, S_s[1]) = \mathrm{Ext}^1_{\mathcal{H}_s}(\Pi_s, S_s) = 0$, the condition ($\star$) is the first non-trivial instance of ($\star\star$).

Theorem 4.8. *Assume the projective case of Setting 4.1, that $\Delta_s, \overline{\nabla}_s \in \mathcal{A}$ for all s (equivalently, condition* (b) *of Proposition 4.6), and that ($\star$) holds. Let $G = \bigoplus_{s=1}^m G_s$ be the iterated universal extension (Definition 2.18) of $(\Delta_1, \ldots, \Delta_m)$ in $\mathcal{A}$.*

1. *G is a projective generator of $\mathcal{A}$, and $\mathrm{Hom}_{\mathcal{D}}(G, Y[2]) = 0$ for all $Y \in \mathcal{A}$. Hence $\mathcal{A} \simeq \mathrm{mod}\,\Lambda$ with $\Lambda = \mathrm{End}_{\mathcal{A}}(G)$.*
2. *With respect to the order $1 < \cdots < m$ of its simple objects $L_1, \ldots, L_m$, the category $\mathcal{A}$ is standardly stratified, with:*
   - *standard objects $\Delta(s) \cong \Delta_s$, and $\mathrm{End}_{\mathcal{A}}(\Delta_s) \cong B_s$;*
   - *proper costandard objects $\overline{\nabla}(s) \cong \overline{\nabla}_s$.*

   It is quasi-hereditary if and only if $B_s \cong \Bbbk$ for all s.
3. *Suppose $\mathcal{D}$ is algebraic. Then* real*: $\mathcal{D}^b(\mathcal{A}) \to \mathcal{D}$ is an equivalence if and only if ($\star\star$) holds. In this case:*
   - *G is a tilting object of $\langle \Delta_1, \ldots, \Delta_m \rangle$, and* real *identifies $K^b(\mathrm{proj}\,\mathcal{A})$ with $\langle \Delta_1, \ldots, \Delta_m \rangle$;*
   - *$\langle \Delta_1, \ldots, \Delta_m \rangle = \mathcal{D}$ if and only if $\mathcal{A}$ is quasi-hereditary;*
   - *each stratum satisfies $\mathcal{T}_s \simeq \mathcal{D}^b(\mathrm{mod}\,B_s)$.*

Proof of (1). We prove by induction on m that the iterated universal extension $G = G^{(m)}$ satisfies:

(i) $\mathrm{Hom}_{\mathcal{D}}(G, Y[l]) = 0$ for all $Y \in \mathcal{A}$ and $l \in \{1, 2\}$;
(ii) $\mathrm{Hom}_{\mathcal{A}}(G, L) \neq 0$ for every simple object L of $\mathcal{A}$.

By Proposition 2.2, (i) says in particular that $\mathrm{Ext}^1_{\mathcal{A}}(G, -) = 0$, so G is projective. By (ii), and since $\mathcal{A}$ is a length category, G is then a projective generator. Hence $\mathcal{A} \simeq \mathrm{mod}\,\mathrm{End}_{\mathcal{A}}(G)$.

Base case $m = 1$. Here $\mathcal{A} = \mathcal{H}_1$ and $G = \Pi_1$. We have $\mathrm{Hom}(\Pi_1, Y[1]) = \mathrm{Ext}^1(\Pi_1, Y) = 0$. The vanishing $\mathrm{Hom}(\Pi_1, Y[2]) = 0$ follows from ($\star$) by dévissage, since all composition factors of Y are isomorphic to S_1. Property (ii) is clear.

Inductive step. Let $m \geq 2$, and write $i = i_m$, $j = j_m$ and $\mathcal{A}' = \mathcal{A}_{m-1}$. For $s < m$ the objects $\Delta_s, \overline{\nabla}_s$ lie in $\mathcal{A}'$ and are the objects of Notation 4.2 for the filtration of $\mathcal{D}_{m-1}$. Hence the induction hypothesis applies to the iterated universal extension $G' = \bigoplus_{s<m} G'_s$ of $(\Delta_1, \dots, \Delta_{m-1})$. For $s < m$ put $V_s := \mathrm{Hom}(G'_s, \Delta_m[1]) = \mathrm{Ext}^1_{\mathcal{A}}(G'_s, \Delta_m)$, which is finite-dimensional. By definition, G_s is given by a short exact sequence, i.e. a triangle

$$\Delta_m \otimes V_s^\vee \xrightarrow{\alpha} G_s \xrightarrow{\beta} G'_s \xrightarrow{\varepsilon} \Delta_m \otimes V_s^\vee[1], \tag{2}$$

where ε is the canonical element. Moreover $G_m = \Delta_m$. We prove (i) and (ii) for G in seven steps.

Step 1: $i^*G_s \cong G'_s$ for $s < m$, and $i^*G_m = 0$. This holds because $i^*\Delta_m = i^*j_!\Pi_m = 0$.

Step 2: $\mathrm{Hom}(G_s, \Delta_m[1]) = 0$ for all s. For $s = m$ this is $\mathrm{Hom}(\Pi_m, \Pi_m[1]) = \mathrm{Ext}^1_{\mathcal{H}_m}(\Pi_m, \Pi_m) = 0$, using that $j_!$ is fully faithful. For $s < m$, apply $\mathrm{Hom}(-, \Delta_m[1])$ to (2). This gives an exact sequence

$$\mathrm{Hom}(\Delta_m \otimes V_s^\vee, \Delta_m) \xrightarrow{\varepsilon^*} \mathrm{Hom}(G'_s, \Delta_m[1]) \to \mathrm{Hom}(G_s, \Delta_m[1]) \to \mathrm{Hom}(\Delta_m \otimes V_s^\vee, \Delta_m[1]).$$

The last term is $\mathrm{End}^1(\Delta_m) \otimes V_s = 0$. The first map is surjective: its restriction to $\Bbbk \cdot \mathrm{id} \otimes V_s$ is the identity of V_s, by the definition of the canonical element.

Step 3: For all s and l we have $\mathrm{Hom}(G_s, \overline{\nabla}_m[l]) \cong V_s \otimes \mathrm{Hom}^l_{\mathcal{T}_m}(\Pi_m, S_m)$ if $s < m$, and $\mathrm{Hom}(G_m, \overline{\nabla}_m[l]) \cong \mathrm{Hom}^l_{\mathcal{T}_m}(\Pi_m, S_m)$. In particular these vanish for $l \in \{1, 2\}$. We have $\mathrm{Hom}^\bullet(G'_s, \overline{\nabla}_m) = 0$, since $G'_s \in \mathcal{D}_{m-1}$ and $\overline{\nabla}_m \in \mathcal{D}^\perp_{m-1}$. So (2) gives $\mathrm{Hom}^\bullet(G_s, \overline{\nabla}_m) \cong V_s \otimes \mathrm{Hom}^\bullet(\Delta_m, \overline{\nabla}_m)$. Now use Lemma 4.4, and for the vanishing use $\mathrm{Ext}^1_{\mathcal{H}_m}(\Pi_m, S_m) = 0$ and ($\star$).

Step 4: $\mathrm{Hom}(G_s, i_*Y[l]) = 0$ for all $Y \in \mathcal{A}'$ and $l \in \{1, 2\}$. By adjunction and Step 1, this group is $\mathrm{Hom}(G'_s, Y[l])$ for $s < m$, which vanishes by the induction hypothesis; for $s = m$ it is 0.

Step 5: $\mathrm{Hom}(G_s, Y[2]) = 0$ for all $Y \in \mathcal{A}$. By dévissage it suffices to treat simple Y. By Proposition 2.13(3) these are the i_*L' with L' simple in $\mathcal{A}'$, which are covered by Step 4, and L_m. Since $\overline{\nabla}_m \in \mathcal{A}$, we have ${}^pj_*S_m = \overline{\nabla}_m$. By Proposition 2.13(2) there is an exact sequence $0 \to L_m \to \overline{\nabla}_m \to C \to 0$ in $\mathcal{A}$ with $C \in i_*\mathcal{A}'$. The associated triangle gives an exact sequence

$$\mathrm{Hom}(G_s, C[1]) \to \mathrm{Hom}(G_s, L_m[2]) \to \mathrm{Hom}(G_s, \overline{\nabla}_m[2]),$$

whose outer terms vanish by Steps 4 and 3.

Step 6: $\mathrm{Hom}(G_s, Y[1]) = 0$ for all $Y \in \mathcal{A}$. Again it suffices to treat $Y = L_m$. By adjunction, $\mathrm{Hom}_{\mathcal{A}}(\Delta_m, L_m) = \mathrm{Hom}(\Pi_m, j^*L_m) = \mathrm{Hom}(\Pi_m, S_m) \neq 0$. Since L_m is simple, we get an exact sequence $0 \to K \to \Delta_m \to L_m \to 0$ in $\mathcal{A}$. The associated triangle gives an exact sequence

$$\mathrm{Hom}(G_s, \Delta_m[1]) \to \mathrm{Hom}(G_s, L_m[1]) \to \mathrm{Hom}(G_s, K[2]),$$

whose outer terms vanish by Steps 2 and 5.

Step 7: property (ii). For L' simple in $\mathcal{A}'$ we have $\mathrm{Hom}(G_s, i_*L') \cong \mathrm{Hom}(G'_s, L')$, which is non-zero for some s by induction. Moreover $\mathrm{Hom}(G_m, L_m) = \mathrm{Hom}(\Delta_m, L_m) \neq 0$.

Steps 5 and 6 give (i), and Step 7 gives (ii). □

Proof of (2). For $s \leq r \leq m$, let $P_r(s)$ be a projective cover of L_s in $\mathcal{A}_r$; this exists by (1) applied to $\mathcal{D}_r$. Recall that $P_{r-1}(s)$ is the largest quotient of $P_r(s)$ lying in the Serre subcategory $\mathcal{A}_{r-1}$. Its kernel is the trace of $P_r(r)$ in $P_r(s)$. We show by induction on $r \geq s$:

$(*)_r$: there is an epimorphism $P_r(s) \twoheadrightarrow \Delta_s$ whose kernel lies in $\mathcal{F}(\Delta_t : s < t \le r)$.

The case $r = s$. By (1) applied to $\mathcal{D}_s$, the object $\Delta_s = G_s^{(s)}$ is projective in $\mathcal{A}_s$. We have $\mathrm{Hom}(\Delta_s, i_{s*}L') = 0$ for $L' \in \mathcal{A}_{s-1}$, and $\mathrm{Hom}(\Delta_s, L_s) \cong \Bbbk$. Hence Δ_s has simple top L_s, so $\Delta_s \cong P_s(s)$. In particular $P_r(r) \cong \Delta_r$ for every r.

The inductive step. Let $r > s$, write $i = i_r$ and $j = j_r$, and let U be the kernel of $P_r(s) \twoheadrightarrow P_{r-1}(s)$, i.e. the trace of Δ_r in $P_r(s)$.

- *j^*U is projective in $\mathcal{H}_r$.* Since $j^*P_{r-1}(s) = 0$, we have $j^*U = j^*P_r(s)$. For $M \in \mathcal{H}_r$ we have $j_*M \in \mathrm{Filt}(\overline{\nabla}_r) \subset \mathcal{A}_r$. Hence
$$\mathrm{Ext}^1_{\mathcal{H}_r}(j^*P_r(s), M) = \mathrm{Hom}(P_r(s), j_*M[1]) = \mathrm{Ext}^1_{\mathcal{A}_r}(P_r(s), j_*M) = 0.$$
So $j^*U \cong \Pi_r^{\oplus q}$ for some q.
- *The counit is an epimorphism.* Consider the counit c: $j_!j^*U \cong \Delta_r^{\oplus q} \to U$; it is a morphism in $\mathcal{A}_r$, and j^*c is an isomorphism. By adjunction, every morphism $\Delta_r \to U$ factors through c. Since U is the sum of the images of all morphisms $\Delta_r \to P_r(s)$, the morphism c is an epimorphism.
- *The counit is a monomorphism.* Let $N = \ker c$. Then $j^*N = 0$, so $N \in i_*\mathcal{A}_{r-1}$. Let $Y \in \mathcal{A}_{r-1}$. We have $\mathrm{Hom}(\Delta_r, i_*Y) = 0$ and $\mathrm{Ext}^1_{\mathcal{A}_r}(\Delta_r, i_*Y) = \mathrm{Hom}(j_!\Pi_r, i_*Y[1]) = 0$. So the sequence $0 \to N \to \Delta_r^{\oplus q} \to U \to 0$ yields
$$\mathrm{Hom}(N, i_*Y) \cong \mathrm{Ext}^1_{\mathcal{A}_r}(U, i_*Y).$$
Since $P_r(s)$ is projective, the sequence $0 \to U \to P_r(s) \to P_{r-1}(s) \to 0$ yields
$$\mathrm{Ext}^1_{\mathcal{A}_r}(U, i_*Y) \cong \mathrm{Ext}^2_{\mathcal{A}_r}(P_{r-1}(s), i_*Y).$$
By Proposition 2.2 the latter group embeds into $\mathrm{Hom}_{\mathcal{D}}(P_{r-1}(s), Y[2])$. This vanishes by (1)(i) applied to $\mathcal{D}_{r-1}$, because $P_{r-1}(s)$ is a direct summand of $G^{(r-1)}$ (Krull–Schmidt). Taking $Y = N$ shows $N = 0$.

Thus $U \cong \Delta_r^{\oplus q}$. Composing $P_r(s) \twoheadrightarrow P_{r-1}(s)$ with the epimorphism of $(*)_{r-1}$ proves $(*)_r$.

Identification of the standard objects. For $r = m$ we obtain $P(s) \twoheadrightarrow \Delta_s$ with kernel $K_s \in \mathcal{F}(\Delta_t : t > s)$. We claim $K_s = U_s := \sum_{t>s} \mathrm{tr}_{P(t)} P(s)$.

- Each Δ_t is a quotient of $P(t)$, so by projectivity every object of $\mathcal{F}(\Delta_t : t > s)$ is a quotient of a direct sum of copies of $P(t)$'s with $t > s$. Hence $K_s \subset U_s$.
- Conversely, $\mathrm{Hom}(P(t), \Delta_s) = 0$ for $t > s$, since $\Delta_s \in \mathcal{A}_s$ has no composition factor L_t. Hence $U_s \subset K_s$.

Therefore $\Delta(s) = P(s)/U_s \cong \Delta_s$ and $U_s \in \mathcal{F}(\Delta(t) : t > s)$, i.e. $\mathcal{A}$ is standardly stratified. Moreover $\mathrm{End}_{\mathcal{A}}(\Delta_s) \cong \mathrm{End}_{\mathcal{T}_s}(\Pi_s) = B_s$, because $j_{s!}$ is fully faithful.

Identification of the proper costandard objects. We have $\overline{\nabla}_s \in \mathcal{A}_s$, and $[\overline{\nabla}_s : L_s]$ equals the length of $j_s^*\overline{\nabla}_s = S_s$, which is 1. We have $\mathrm{Hom}(i_{s*}L', \overline{\nabla}_s) = \mathrm{Hom}(L', i_s^!j_{s*}S_s) = 0$ for L' simple in $\mathcal{A}_{s-1}$, so $\operatorname{soc}\overline{\nabla}_s = L_s$ and $\overline{\nabla}_s \hookrightarrow I(s)$. Furthermore $\mathrm{Ext}^1_{\mathcal{A}}(i_{s*}L', \overline{\nabla}_s) = \mathrm{Hom}(L', i_s^!j_{s*}S_s[1]) = 0$. Hence no subobject of $I(s)$ properly containing $\overline{\nabla}_s$ has composition factors L_t with $t < s$ besides a single L_s: such a subobject M would contain a non-split extension of some L_t ($t < s$) by $\overline{\nabla}_s$. Thus $\overline{\nabla}_s = \overline{\nabla}(s)$.

The last assertion of (2) is Lemma 2.27(2). □

Proof of (3). *Sufficiency of* ($\star\star$). Assume ($\star\star$). Repeat the proof of (1) with $l \in \{1, 2\}$ replaced by all $l \ge 1$:

- in the base case and in Step 3, ($\star\star$) gives the vanishing for all $l \ge 1$;
- Step 4 holds for all $l \ge 1$ by induction;

- in Step 5, for every $l \geq 2$, the exact sequence $\mathrm{Hom}(G_s, C[l-1]) \to \mathrm{Hom}(G_s, L_m[l]) \to \mathrm{Hom}(G_s, \overline{\nabla}_m[l])$ has vanishing outer terms;
- Step 6 is unchanged.

Hence $\mathrm{Hom}_{\mathcal{D}}(G, Y[l]) = 0$ for all $Y \in \mathcal{A}$ and $l \geq 1$. Let $X, Y \in \mathcal{A}$, $n \geq 2$ and $f \in \mathrm{Hom}(X, Y[n])$. Choose an epimorphism $G^{\oplus r} \twoheadrightarrow X$; its composite with f vanishes. By Theorem 2.3, real is fully faithful, hence an equivalence.

Consequences. Then $G \in \mathcal{A}$ satisfies $\mathrm{Hom}(G, G[l]) = 0$ for all $l \neq 0$. By induction, using the triangles (2) and $G_m = \Delta_m$, we have $\langle G \rangle = \langle \Delta_1, \dots, \Delta_m \rangle$. Since real is the identity on $\mathcal{A}$, it maps $K^b(\mathrm{proj}\,\mathcal{A}) = \langle G \rangle_{\mathcal{D}^b(\mathcal{A})}$ onto $\langle G \rangle_{\mathcal{D}}$. Equality $\langle \Delta_1, \dots, \Delta_m \rangle = \mathcal{D}$ thus means $K^b(\mathrm{proj}\,\mathcal{A}) = \mathcal{D}^b(\mathcal{A})$, i.e. $\mathrm{gl.dim}\,\mathcal{A} < \infty$; by Theorem 6.3(4) this is equivalent to quasi-heredity. The equivalence $\mathcal{T}_s \simeq \mathcal{D}^b(\mathrm{mod}\,B_s)$ is the case $m = 1$ applied to $\mathcal{T}_s$, which is algebraic, being equivalent to a thick subcategory of $\mathcal{D}$.

Necessity of ($\star\star$). Conversely, suppose real is an equivalence. Then, using Lemma 4.4, (2) and Lemma 2.27(3),

$$\mathrm{Hom}^l_{\mathcal{T}_s}(\Pi_s, S_s) \cong \mathrm{Hom}^l_{\mathcal{D}}(\Delta_s, \overline{\nabla}_s) \cong \mathrm{Ext}^l_{\mathcal{A}}(\Delta(s), \overline{\nabla}(s)) = 0 \qquad \text{for } l \geq 1.$$

This is ($\star\star$). □

Corollary 4.9 (cf. Theorem 2.21(2) and Theorem 2.22)**.** *Let $\mathcal{A}$ be an abelian category. Suppose $\mathcal{D}^b(\mathcal{A})$ admits an admissible filtration as in the projective case of Setting 4.1 such that:*

- *the t-structures on the strata are bounded with local hearts;*
- *$\Delta_s, \overline{\nabla}_s \in \mathcal{A}$ for all s;*
- *($\star$) holds.*

Then $\mathcal{A}$ is standardly stratified, with standard objects Δ_s and proper costandard objects $\overline{\nabla}_s$. Moreover ($\star\star$) holds, and $\mathcal{T}_s \simeq \mathcal{D}^b(\mathrm{mod}\,B_s)$ for all s.

Proof. By Proposition 4.7, the standard t-structure of $\mathcal{D}^b(\mathcal{A})$ is the glued one, so Theorem 4.8(1),(2) apply. The identity functor of $\mathcal{D}^b(\mathcal{A})$ is a realisation functor for the standard t-structure, so Theorem 4.8(3) gives the remaining statements. □

4.4. **Comparison with known results.**

Remark 4.10 (Bodzenta–Bondal). In the exceptional case (Example 4.3(1)) we have $\Pi_s = S_s = \Bbbk$, $B_s = \Bbbk$, and ($\star\star$) holds trivially. Then:

- Proposition 4.6 together with Theorem 4.8(1),(2) is Theorem 2.20(1). Indeed, standardly stratified with $B_s = \Bbbk$ means quasi-hereditary, i.e. highest weight, and for quasi-hereditary algebras $\overline{\nabla} = \nabla$.
- Theorem 4.8(3) is Theorem 2.20(2), with the hypothesis "dg enhanced and idempotent complete" replaced by "algebraic".
- Proposition 4.5 is Theorem 2.16, and Proposition 4.7 and Corollary 4.9 are the corresponding parts of Theorems 2.21 and 2.22.

The proof of Theorem 4.8(1) differs from that of [BB26]. Bodzenta and Bondal first show that G is a tilting generator of $\mathcal{D}$ and then compare t-structures. When $B_s \neq \Bbbk$, the object G generates only $\langle \Delta_1, \dots, \Delta_m \rangle \subsetneq \mathcal{D}$ (Theorem 6.3), so this route is unavailable. Instead we prove Ext-projectivity directly, in degrees 1 and 2. This is the reason why the degree-two condition ($\star$) appears.

Remark 4.11 (Stratifying systems). The sequence $(\Delta_1, \dots, \Delta_m)$ in $\mathcal{A}$ satisfies the following conditions: $\mathrm{Hom}(\Delta_t, \Delta_s) = 0$ for $t > s$, $\mathrm{Ext}^1(\Delta_t, \Delta_s) = 0$ for $t \geq s$, and each $\mathrm{End}(\Delta_s)$ is local. These are the axioms of a *stratifying system* in the sense of Erdmann and Sáenz [ES03]; see

also [MMS04]. The theory of stratifying systems generalises the Dlab–Ringel standardisation theorem (Theorem 2.19) from quasi-hereditary to standardly stratified algebras, by passing to the endomorphism algebra of a relative projective generator. Theorem 4.8(1),(2) says that in the glued situation no such passage is needed, in the same way that Theorem 2.20 sharpens Theorem 2.19.

Remark 4.12 (On the role of ($\star$)). In the proof of Theorem 4.8(1), condition ($\star$) for the top stratum $s = m$ enters only through Step 3, via the tensor products $V_s \otimes \mathrm{Hom}^2(\Pi_m, S_m)$. It can therefore be dropped for $s = m$ if $V_s = \mathrm{Ext}^1_{\mathcal{A}}(G'_s, \Delta_m) = 0$ for all $s < m$. Condition ($\star$) for $s < m$ enters through the vanishing of $\mathrm{Hom}(G', \mathcal{A}'[2])$, which is used in Steps 4 and 5 and in the proof of (2). Example 5.5 shows that the conclusion of Theorem 4.8(1) may hold even though ($\star$) fails.

5. Semi-orthogonal sequences of $\mathbb{P}^n[k]$-like objects

Definition 5.1. A $\mathbb{P}$-*sequence* in $\mathcal{D}$ is a sequence $(P_1, \dots, P_m)$ of objects with the following properties:

- each P_s is $\mathbb{P}^{n_s}[k_s]$-like for some $n_s \geq 0$ and some $k_s \geq 1$ (for $n_s = 0$ the value of k_s plays no role);
- $\mathrm{Hom}^\bullet(P_t, P_s) = 0$ for $t > s$.

Put $\mathcal{D}_s := \langle P_1, \dots, P_s \rangle$. The sequence is *full* if $\mathcal{D}_m = \mathcal{D}$, and *admissible* if $\mathcal{D}_{s-1}$ is admissible in $\mathcal{D}_s$ for all s.

If all $n_s = 0$, a $\mathbb{P}$-sequence is an exceptional sequence, and admissibility holds automatically under mild finiteness assumptions [Bon89]. For $n_s \geq 1$ the subcategory $\langle P_s \rangle \simeq \mathrm{perf}(A_{n_s,k_s})$ is proper but not of the form $\mathcal{D}^b(\Bbbk)$. We do not know a general criterion for admissibility, and therefore impose it as a hypothesis.

Intuition. A $\mathbb{P}$-sequence is to a $\mathbb{P}$-like object what an exceptional sequence is to an exceptional object: an admissible filtration all of whose strata are generated by a single $\mathbb{P}$-like object. Since every stratum carries, up to shift, a unique non-trivial t-structure (Remark 3.13), there is essentially one way to glue, and the question is what the resulting heart looks like.

Lemma 5.2. *Let $(P_1, \dots, P_m)$ be a full admissible $\mathbb{P}$-sequence in an algebraic triangulated category $\mathcal{D}$.*

1. *The subcategory $\langle P_s \rangle$ equals ${}^{\perp}\mathcal{D}_{s-1} \cap \mathcal{D}_s$, and the quotient functor induces an equivalence $\langle P_s \rangle \simeq \mathcal{T}_s$. In particular $\mathcal{T}_s \simeq \mathrm{perf}(A_{n_s,k_s})$, and $j_{s!}$ is the inclusion $\langle P_s \rangle \subset \mathcal{D}_s$.*
2. *Put $E_s := P_s = j_{s!}P_s$ and $F_s := j_{s*}P_s$. Then F_s is the cone of the counit $i_{s*}i_s^! P_s \to P_s$, i.e. the left mutation of P_s through $\mathcal{D}_{s-1}$. If all $n_s = 0$, then $(F_m, \dots, F_1)$ is the left dual exceptional sequence.*
3. $\mathrm{Hom}^\bullet(E_s, F_t) \cong \delta_{st}\, \mathrm{End}^\bullet(P_s)$.

Proof. (1) By admissibility, the quotient functor j_s^* restricts to an equivalence ${}^{\perp}\mathcal{D}_{s-1} \cap \mathcal{D}_s \simeq \mathcal{T}_s$. The subcategory ${}^{\perp}\mathcal{D}_{s-1} \cap \mathcal{D}_s$ is thick and contains P_s, hence contains $\langle P_s \rangle$. The restriction of j_s^* to $\langle P_s \rangle$ is fully faithful. Its essential image is thick, because $\langle P_s \rangle$ is idempotent complete, and it contains $j_s^* P_s$. Since $\mathcal{D}_s$ is generated by $\mathcal{D}_{s-1}$ and P_s, the object $j_s^* P_s$ generates $\mathcal{T}_s$. Hence the restriction is an equivalence, and $\langle P_s \rangle = {}^{\perp}\mathcal{D}_{s-1} \cap \mathcal{D}_s$. The model $\mathrm{perf}(A_{n_s,k_s})$ comes from Theorem 3.5.

(2) The first claim is the triangle $i_* i^! X \to X \to j_* j^* X$ of Definition 2.11 for $X = P_s$. For exceptional sequences this is the description $F_s = j_{s*}(\Bbbk)$ of the left dual recalled in Section 2.

(3) This is Lemma 4.4 with the test objects $Q_s = S_s = P_s$. □

We equip each $\mathcal{T}_s = \langle P_s \rangle$ with the t-structure t_{P_s} of Proposition 3.12. By Remark 3.13, every t-structure on $\mathcal{D}$ glued from non-trivial t-structures on the strata is obtained from this one by shifting the t-structure on each stratum by some integer $p(s)$ (a "perversity"). Replacing P_s by $P_s[p(s)]$ does not change the filtration, so it suffices to treat the unshifted case. Let $\mathcal{A}$ denote the glued heart.

Proposition 5.3 (Criterion). *$E_s, F_s \in \mathcal{A}$ for all s if and only if $\operatorname{Hom}(E_s, E_t[l]) = 0$ and $\operatorname{Hom}(F_s, F_t[l]) = 0$ for all s, t and all $l < 0$.*

Proof. This is Proposition 4.6 with the test objects $Q_s = S_s = P_s$; note that $\operatorname{End}^\bullet(P_s)$ is concentrated in non-negative degrees. □

Theorem 5.4. *Let $(P_1, \dots, P_m)$ be a full admissible $\mathbb{P}$-sequence in an algebraic triangulated category $\mathcal{D}$, and assume $E_s, F_s \in \mathcal{A}$ for all s.*

1. *Suppose that for every s either $n_s = 0$ or $k_s \geq 3$. Then $\mathcal{A}$ is a highest weight category with respect to $1 < \cdots < m$, with standard objects E_s and costandard objects F_s.*
2. *Suppose that for every s one of the following holds: $n_s = 0$; or $k_s \geq 3$; or $k_s = 1$ and $n_s \geq 3$. Then $\mathcal{A} \simeq \operatorname{mod} \Lambda$ with Λ standardly stratified with respect to $1 < \cdots < m$, and:*
   - *$B_s \cong \Bbbk$ in the first two cases, and $B_s \cong \Bbbk[x]/(x^2)$ in the third case;*
   - *the standard objects are $\Delta(s) = E_s$ in the first two cases, and $\Delta(s) = j_{s!}Q_s$ in the third case, where $Q_s \in \langle P_s \rangle$ is the object of Proposition 3.14(3); in the third case there is an exact sequence $0 \to E_s \to \Delta(s) \to E_s \to 0$ in $\mathcal{A}$;*
   - *the proper standard objects are $\overline{\Delta}(s) \cong E_s$, and the proper costandard objects are $\overline{\nabla}(s) \cong F_s$.*
3. *In the situations of (1) and (2), real: $\mathcal{D}^b(\mathcal{A}) \to \mathcal{D}$ is an equivalence if and only if $n_s = 0$ for all s.*
4. *If $k_s = n_s = 1$ for some s, then the simple object L_s of $\mathcal{A}$ has no projective cover. In particular $\mathcal{A}$ is not equivalent to $\operatorname{mod} \Lambda$ for any finite-dimensional algebra Λ.*

Proof. Reduction to Theorem 4.8. In the situations of (1) and (2) we are in the projective case of Setting 4.1, with $S_s = P_s$. Moreover:

- if $n_s = 0$ or $k_s \geq 2$, then $\Pi_s = P_s$ and $B_s = \Bbbk$ (Proposition 3.14(1));
- if $k_s = 1$ and $n_s \geq 2$, then $\Pi_s = Q_s$ and $B_s \cong \Bbbk[x]/(x^2)$ (Proposition 3.14(3)).

Condition ($\star$) holds:

- for $n_s = 0$ trivially;
- for $k_s \geq 3$ because $\operatorname{End}^\bullet(P_s)$ is concentrated in degrees divisible by k_s;
- for $k_s = 1$ and $n_s \geq 3$ because $\operatorname{Hom}(Q_s, P_s[l]) \neq 0$ only for $l \in \{0, n_s\}$.

Since $\Pi_s \in \operatorname{Filt}(P_s)$, we have $\Delta_s = j_{s!}\Pi_s \in \operatorname{Filt}(E_s) \subset \mathcal{A}$, and $\overline{\nabla}_s = F_s$. Hence Theorem 4.8 applies.

Proof of (2). Everything except the statement about $\overline{\Delta}(s)$ in the third case is Theorem 4.8(1),(2). In the third case, applying $j_{s!}$ to $0 \to P_s \to Q_s \to P_s \to 0$ gives a triangle $E_s \to \Delta(s) \to E_s \to$ with all vertices in $\mathcal{A}$, i.e. an exact sequence. We have $[\Delta(s) : L_s] = 2$, since $j_s^*\Delta(s) = Q_s$ has length two, and $[E_s : L_s] = 1$. Let $N \subset \Delta(s)$ be a subobject with $[\Delta(s)/N : L_s] = 1$ and $N \subset E_s$, where $E_s \subset \Delta(s)$ is the subobject above. Then E_s/N has no composition factor L_s, so it lies in $i_{s*}\mathcal{A}_{s-1}$. But $E_s = j_{s!}P_s = {}^p j_{s!}P_s$ has no non-zero quotient in $i_{s*}\mathcal{A}_{s-1}$ (Proposition 2.13(2)), so $N = E_s$. Since every quotient of $\Delta(s)$ with a single composition factor L_s factors through $\Delta(s)/E_s$ (because $\operatorname{rad}\operatorname{End}(\Delta(s))$ maps $\Delta(s)$ onto the subobject E_s), we conclude $\overline{\Delta}(s) = \Delta(s)/E_s \cong E_s$.

Proof of (1). We have $B_s = \Bbbk$ for all s, so $\mathcal{A}$ is quasi-hereditary, i.e. highest weight. For quasi-hereditary algebras $\nabla(s) = \overline{\nabla}(s)$, so the costandard objects are the F_s.

Proof of (3). By Theorem 4.8(3) it suffices to check when ($\star\star$) holds. If $n_s \geq 1$ and $k_s \geq 3$, then $\mathrm{Hom}(P_s, P_s[k_s]) \neq 0$. If $k_s = 1$ and $n_s \geq 3$, then $\mathrm{Hom}(Q_s, P_s[n_s]) \neq 0$. If $n_s = 0$, the condition holds.

Proof of (4). Suppose R is a projective cover of L_s in $\mathcal{A}$, and let $\bar{R}$ be its largest quotient lying in the Serre subcategory $\mathcal{A}_s$; then $\bar{R}$ is a projective cover of L_s in $\mathcal{A}_s$.

- For $M \in \mathcal{H}_s$ we have $j_{s*}M \in \mathrm{Filt}(F_s) \subset \mathcal{A}_s$. Hence
$$\mathrm{Ext}^1_{\mathcal{H}_s}(j_s^*\bar{R}, M) = \mathrm{Hom}(\bar{R}, j_{s*}M[1]) = \mathrm{Ext}^1_{\mathcal{A}_s}(\bar{R}, j_{s*}M) = 0,$$
so $j_s^*\bar{R}$ is projective in $\mathcal{H}_s$.
- $j_s^*\bar{R} \neq 0$: otherwise $\bar{R} \in i_{s*}\mathcal{A}_{s-1}$, and then $\mathrm{Hom}(\bar{R}, L_s) = 0$ because $L_s = j_{s!*}S_s$ has no non-zero subobject in $i_{s*}\mathcal{A}_{s-1}$.

This contradicts Proposition 3.14(2). □

Example 5.5 (A case with $k = 2$)**.** Let Γ be the graded $\Bbbk$-linear category (a dg category with zero differential) with two objects $1, 2$ and the following morphism spaces:

- $\Gamma(1,1) = \Bbbk$ and $\Gamma(2,2) = \Bbbk[t]/(t^2)$ with $\deg t = 2$;
- $\Gamma(1,2) = \Bbbk f$ with $\deg f = 0$, and $\Gamma(2,1) = 0$;
- $t \circ f = 0$.

Let $\mathcal{D} = \mathrm{perf}(\Gamma)$, and let P_1, P_2 be the representable objects, so that $\mathrm{Hom}^\bullet(P_i, P_j) = \Gamma(i,j)$. Then (P_1, P_2) is a full $\mathbb{P}$-sequence with $n_1 = 0$ and with P_2 a $\mathbb{P}^1[2]$-like object. It is admissible, because $\langle P_1 \rangle \simeq \mathcal{D}^b(\Bbbk)$ is generated by an exceptional object and $\mathcal{D}$ is proper.

We have $i^!P_2 = P_1$, hence $F_2 = \mathrm{cone}(P_1 \xrightarrow{f} P_2)$ and $F_1 = E_1 = P_1$. Checking the negative Homs:

- $\mathrm{Hom}^\bullet(E_1, E_2) = \Bbbk f$ is concentrated in degree 0;
- $\mathrm{Hom}^\bullet(F_1, F_2) = 0$;
- $\mathrm{Hom}^l(F_2, F_1) \cong \mathrm{End}^{l-1}(P_1)$ is concentrated in degree 1;
- $\mathrm{End}^\bullet(F_2) \cong \mathrm{End}^\bullet(P_2)$.

By Proposition 5.3, $E_s, F_s \in \mathcal{A}$ for $s = 1, 2$.

The triangle $P_1 \to P_2 \to F_2 \to P_1[1]$ has all vertices in $\mathcal{A}$, so it gives an exact sequence $0 \to E_1 \to E_2 \to F_2 \to 0$. Hence the simple objects are $L_1 = E_1$ and $L_2 = F_2$. Their extensions are:

- $\mathrm{Ext}^1_{\mathcal{A}}(L_2, L_1) = \mathrm{Hom}(F_2, P_1[1]) \cong \Bbbk$;
- $\mathrm{Ext}^1_{\mathcal{A}}(L_1, L_2) = \mathrm{Hom}(P_1, F_2[1]) = 0$;
- $\mathrm{Ext}^1_{\mathcal{A}}(L_1, L_1) = \mathrm{Ext}^1_{\mathcal{A}}(L_2, L_2) = 0$.

Moreover $\mathrm{Ext}^1_{\mathcal{A}}(E_1, -)$ and $\mathrm{Ext}^1_{\mathcal{A}}(E_2, -)$ vanish on both simple objects, so E_1, E_2 are the indecomposable projective objects. Thus $\mathcal{A} \simeq \mathrm{mod}\, \Bbbk\vec{A}_2$ is highest weight with standard objects E_1, E_2 and costandard objects F_1, F_2: the conclusion of Theorem 5.4(1) holds even though $k_2 = 2$.

This does not contradict the hypotheses of Theorem 5.4: here $V_1 = \mathrm{Ext}^1_{\mathcal{A}}(E_1, E_2) = 0$, so by Remark 4.12 condition ($\star$) is not needed for the top stratum. The realisation functor is not an equivalence, since $\mathrm{Hom}(P_2, P_2[2]) \neq 0$ while $\mathrm{Ext}^2_{\Bbbk\vec{A}_2} = 0$.

Remark 5.6 (The obstruction for $k = 2$). Suppose all strata have semisimple hearts, i.e. $k_s \geq 2$ or $n_s = 0$. In the proof of Theorem 4.8(1), the only point where ($\star$) is needed for a stratum $t < m$ is Step 6. There the kernel K of $\Delta_m \twoheadrightarrow L_m$ lies in $i_*\mathcal{A}_{m-1}$, and one needs $\mathrm{Hom}(G_s, K[2]) = 0$. If some $k_t = 2$ with $t < m$, this group can be non-zero, and whether $\mathrm{Ext}^1_{\mathcal{A}}(G_s, L_m)$ vanishes then depends on the connecting map $\mathrm{Hom}(G_s, L_m[1]) \to \mathrm{Hom}(G_s, K[2])$. This map is determined by

higher (Massey-type) composition data and not by the graded endomorphism algebras alone; compare the non-formality of the Koszul duals in Remark 3.7.

Question 5.7. *Let $(P_1, \dots, P_m)$ be a full admissible $\mathbb{P}$-sequence with $E_s, F_s \in \mathcal{A}$, and suppose some $k_s = 2$, or some $(k_s, n_s) = (1, 2)$. Is $\mathcal{A}$ Deligne finite? If so, is it standardly stratified with standard objects $j_{s!}\Pi_s$, as in Theorem 5.4?*

A natural source of test cases is geometry:

- the structure sheaves of hyperkähler manifolds are $\mathbb{P}^n[2]$-objects;
- by [BC25, Thm. B], as quoted in [Bonf26], all indecomposable perverse sheaves on $\mathbb{P}^n$ are $\mathbb{P}[2]$-like, and most of them are not Calabi–Yau.

It would be interesting to know whether the latter can be arranged into $\mathbb{P}$-sequences satisfying the hypotheses above.

6. Stratified algebras and local-exceptional sequences

Definition 6.1. An object E is *local-exceptional* if $\mathrm{Hom}(E, E[l]) = 0$ for all $l \neq 0$ and $\mathrm{End}(E)$ is a (finite-dimensional) local algebra. A *local-exceptional sequence* is a sequence $(E_1, \dots, E_m)$ of local-exceptional objects with $\mathrm{Hom}^\bullet(E_t, E_s) = 0$ for $t > s$; it is *full* if $\langle E_1, \dots, E_m \rangle = \mathcal{D}$.

Intuition. For a local-exceptional object E with $B = \mathrm{End}(E)$, the subcategory $\langle E \rangle$ is equivalent to $K^b(\mathrm{proj}\, B)$ (Theorem 2.10), and E plays the role of the free module. $\mathbb{P}^n[0]$-like objects are the case $B \cong \Bbbk[t]/(t^{n+1})$. By Theorem 3.6(3), such strata carry weight structures but no non-trivial t-structures. The "dual" of E is the simple B-module, which lives in $\mathcal{D}^b(\mathrm{mod}\, B)$ but not in $K^b(\mathrm{proj}\, B)$ unless $B = \Bbbk$.

Lemma 6.2. *Let $\mathcal{C} = \langle \mathcal{U} \cup \mathcal{V} \rangle$ for thick subcategories $\mathcal{U}, \mathcal{V}$ with $\mathrm{Hom}^\bullet(\mathcal{V}, \mathcal{U}) = 0$.*

1. *Every $X \in \mathcal{C}$ fits into a triangle $V_X \to X \to U_X \to V_X[1]$ with $V_X \in \mathcal{V}$ and $U_X \in \mathcal{U}$.*
2. *If $\mathrm{Hom}^\bullet(\mathcal{V}, X) = 0$, then $X \in \mathcal{U}$.*

Proof. (1) Since $\mathrm{Hom}^\bullet(\mathcal{V}, \mathcal{U}) = 0$, the morphism $V_X \to X$ in such a triangle is a right $\mathcal{V}$-approximation inducing $\mathrm{Hom}(V, V_X) \cong \mathrm{Hom}(V, X)$ for all $V \in \mathcal{V}$. Hence the triangle is unique and functorial. The objects admitting such a triangle form a triangulated subcategory (see e.g. [Bon89, BK89]). By functoriality this subcategory is closed under direct summands, so it is thick. It contains $\mathcal{U}$ and $\mathcal{V}$, so it equals $\mathcal{C}$.

(2) If $\mathrm{Hom}^\bullet(\mathcal{V}, X) = 0$, the morphism $V_X \to X$ vanishes, so $U_X \cong X \oplus V_X[1]$. Then $V_X[1]$ is a direct summand of U_X, hence lies in $\mathcal{U} \cap \mathcal{V}$. This intersection is 0 because $\mathrm{Hom}(\mathcal{V}, \mathcal{U}) = 0$. Hence $X \cong U_X \in \mathcal{U}$. □

Theorem 6.3. *Let Λ be a standardly stratified algebra with respect to $1 < \cdots < m$, and let $B_i = \mathrm{End}(\Delta(i))$.*

1. *$(\Delta(1), \dots, \Delta(m))$ is a full local-exceptional sequence in $K^b(\mathrm{proj}\, \Lambda)$ with $\mathrm{End}(\Delta(i)) = B_i$. In particular, if $B_i \cong \Bbbk[t]/(t^{n+1})$, then $\Delta(i)$ is $\mathbb{P}^n[0]$-like.*
2. *In $\mathcal{D}^b(\mathrm{mod}\, \Lambda)$ we have $\mathrm{Hom}^\bullet(\Delta(i), \overline{\nabla}(i)) \cong \Bbbk$ (in degree 0) and $\mathrm{Hom}^\bullet(\Delta(j), \overline{\nabla}(i)) = 0$ for $j > i$.*
3. *If $B_i \neq \Bbbk$, then $\overline{\nabla}(i) \notin K^b(\mathrm{proj}\, \Lambda)$, i.e. $\mathrm{pd}\, \overline{\nabla}(i) = \infty$.*
4. *$\mathrm{gl.dim}\, \Lambda < \infty$ if and only if Λ is quasi-hereditary, if and only if $B_i \cong \Bbbk$ for all i.*

Proof. (1) and (2) are Lemma 2.27.

(3) Suppose $X := \overline{\nabla}(i) \in K^b(\mathrm{proj}\, \Lambda) = \langle \Delta(1), \dots, \Delta(m) \rangle$.

- Apply Lemma 6.2 with $\mathcal{U} = \langle \Delta(1), \dots, \Delta(i) \rangle$ and $\mathcal{V} = \langle \Delta(i+1), \dots, \Delta(m) \rangle$. Since $\mathrm{Hom}^\bullet(\mathcal{V}, X) = 0$ by (2), we get $X \in \mathcal{U}$.

- Apply Lemma 6.2(1) to $\mathcal{U} = \langle \mathcal{U}' \cup \mathcal{V}' \rangle$ with $\mathcal{U}' = \langle \Delta(1), \dots, \Delta(i-1) \rangle$ and $\mathcal{V}' = \langle \Delta(i) \rangle$. This gives a triangle $V \to X \to U \to V[1]$ with $V \in \langle \Delta(i) \rangle$ and $U \in \mathcal{U}'$. Since $\mathrm{Hom}^\bullet(\Delta(i), U) = 0$, we get $\mathrm{Hom}^\bullet(\Delta(i), V) \cong \mathrm{Hom}^\bullet(\Delta(i), X) \cong \Bbbk$.
- The functor $\Phi := \mathbf{R}\mathrm{Hom}_\Lambda(\Delta(i), -) \colon \mathcal{D}^b(\operatorname{mod}\Lambda) \to \mathcal{D}(\operatorname{Mod} B_i)$ is exact and sends $\Delta(i)$ to B_i, because $\Delta(i)$ has no self-extensions. Hence $\Phi(V) \in \langle B_i \rangle = K^b(\operatorname{proj} B_i)$, with total cohomology $\mathrm{Hom}^\bullet(\Delta(i), V) \cong \Bbbk$.

By Lemma 3.8, $\dim B_i$ divides 1, so $B_i = \Bbbk$.

(4) If $\operatorname{gl.dim}\Lambda < \infty$, every module is perfect, so $B_i = \Bbbk$ for all i by (3). Then Λ is quasi-hereditary by Lemma 2.27(2). Conversely, quasi-hereditary algebras have finite global dimension [CPS88, DR92]. □

Remark 6.4.

(1) Statement (4) is presumably well known to experts (cf. [AHLU00, Dla96]). We include the short Euler-characteristic argument because it isolates exactly the obstruction to a "single smooth category" version of Theorem 2.20. In a genuinely stratified situation, the standard objects generate only $K^b(\operatorname{proj}\Lambda)$, while their Hom-dual partners $\overline{\nabla}(i)$ lie outside it. This is why Theorem 4.8 is formulated for strata of "$\mathcal{D}^b$-type", which contain both Π_s and S_s.

(2) For $B_i = \Bbbk[t]/(t^{n+1})$, the self-extension algebra of $\overline{\nabla}(i)$ within its stratum is the Koszul dual $\mathrm{Ext}^\bullet_{B_i}(\Bbbk, \Bbbk)$ (Remark 3.7 with $k = 0$). It has cohomology $\Bbbk[s]$ with $\deg s = 1$ if $n = 1$, and $\Bbbk[s,t]/(s^2)$ with $\deg s = 1$, $\deg t = 2$ if $n > 1$. Thus the standard objects are $\mathbb{P}^n[0]$-like, while the proper costandard objects are not $\mathbb{P}$-like for $n \geq 2$; they are "$\mathbb{P}^\infty[1]$-like" for $n = 1$.

Question 6.5 (Converse of Corollary 4.9)**.** *Let Λ be standardly stratified. Does $\mathcal{D}^b(\operatorname{mod}\Lambda)$ admit an admissible filtration with strata $\mathcal{T}_s \simeq \mathcal{D}^b(\operatorname{mod} B_s)$ whose glued t-structure is the standard one, and such that $\Delta_s = \Delta(s)$ and $\overline{\nabla}_s = \overline{\nabla}(s)$?*

The natural candidate is $\mathcal{D}_s := \mathcal{D}^b_{\mathcal{A}_s}(\operatorname{mod}\Lambda)$, the complexes with cohomology in the Serre subcategory $\mathcal{A}_s$ spanned by $L(1), \dots, L(s)$. For quasi-hereditary algebras this works, by [CPS88] and [BB26, Cor. 3.6]. For standardly stratified algebras, the idempotent ideals defining the $\mathcal{A}_s$ are projective only as one-sided modules. This suggests that in general only one of the two adjoints required for a recollement of bounded derived categories exists. We expect a positive answer for properly stratified algebras, and a version of Theorem 4.8 with one-sided gluing in general. Together with Corollary 4.9, a positive answer would give an analogue of Theorem 2.22: standardly stratified structures on $\mathcal{A}$ would correspond to admissible filtrations of $\mathcal{D}^b(\mathcal{A})$ with local strata such that the objects Δ_s and $\overline{\nabla}_s$ lie in $\mathcal{A}$.

7. Further questions

Question 7.1 (Classification of glued t-structures)**.** *For a full admissible $\mathbb{P}$-sequence with all $k_s \geq 1$, Theorem 3.6 implies that the t-structures on $\mathcal{D}$ glued along the filtration from non-trivial t-structures on the strata are parametrised by perversity functions $p \colon \{1, \dots, m\} \to \mathbb{Z}$. Which t-structures on $\mathcal{D}$ are glued in this sense? For which perversities do the objects $E_s[p(s)]$ and $F_s[p(s)]$ lie in the heart, as in the middle perversity of* [BB26, §5.1]*?*

Question 7.2 (Ringel duality)**.** *By* [BB26, Thm. 3.2]*, the Ringel dual of a highest weight category glued along an exceptional sequence is glued along the left dual sequence. Characteristic tilting modules and Ringel duality are available for standardly and properly stratified algebras (see e.g.* [AHLU00]*). In Setting 4.1, which admissible filtration of $\mathcal{D}$, and which choice of objects in the strata, produces the Ringel dual of the heart in Theorem 4.8?*

Question 7.3 (The weight-structure side)**.** *In the situation of Theorem 4.8(3), the subcategory* $\langle \Delta_1, \dots, \Delta_m \rangle \simeq K^b(\operatorname{proj} \Lambda)$ *is filtered with strata* $\langle \Delta_s \rangle \simeq K^b(\operatorname{proj} B_s)$*. These strata carry the weight structures with co-hearts* $\operatorname{add} \Delta_s$*, which are unique up to shift for* $B_s = \Bbbk[t]/(t^{n+1})$ *by Theorem 3.6(3). Is the co-t-structure obtained by gluing these* [LVY14] *the one with co-heart* $\operatorname{add} G$*? Does it correspond to the glued t-structure under the Koenig–Yang bijection (Theorem 2.9)? A positive answer would give a "silting" counterpart of Theorem 4.8, with* $\mathbb{P}^n[0]$*-like strata.*

Question 7.4 (Infinite n)**.** *Remark 3.16 shows that* $\mathcal{D}^b(\operatorname{mod} \Bbbk[x]/(x^2))$ *is generated by a "*$\mathbb{P}^\infty[1]$*-like" object. Is there a natural class of "*$\mathbb{P}^\infty[k]$*-like" strata (graded endomorphism algebra* $\Bbbk[t]$*) for which Theorem 4.8(3) applies, so that* $\mathbb{P}^n[1]$*-like strata are their truncations?*

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